RESEARCH & RELATED PERSONAL DATA
Project Director/Principal Investigator and Co-Project Director(s)/Co-Principal Investigator(s)

The Federal Government has a continuing commitment to monitor the operation of its review and award processes to identify and address any inequities based on gender, race, ethnicity, or disability of its proposed PDs/PIs and co-PDs/PIs. To gather information needed for this important task, the applicant should submit the requested information for each identified PD/PI and co-PDs/PIs with each proposal. Submission of the requested information is voluntary and is not a precondition of award. However, information not submitted will seriously undermine the statistical validity, and therefore the usefulness, of information received from others. Any individual not wishing to submit some or all the information should check the box provided for this purpose. Upon receipt of the application, this form will be separated from the application. This form will not be duplicated, and it will not be a part of the review process. Data will be confidential.

Project Director/Principal Investigator

Prefix:	*** First Name:**	**Middle Name:**	*** Last Name:**	**Suffix:**
	David		Molik	

Gender: Male

Race (check all that apply):
❍ American Indian or Alaska Native
❍ Asian
❍ Black or African American
❍ Native Hawaiian or Other Pacific Islander
● White
❍ Do Not Wish to Provide

Ethnicity:
Do Not Wish To Provide

Disability Status (check all that apply):
❍ Hearing
❍ Visual
❍ Mobility/Orthopedic Impairment
❍ Other
❍ None
● Do Not Wish to Provide

Citizenship:
US Citizen

RESEARCH & RELATED PERSONAL DATA
Project Director/Principal Investigator and Co-Project Director(s)/Co-Principal Investigator(s)

The Federal Government has a continuing commitment to monitor the operation of its review and award processes to identify and address any inequities based on gender, race, ethnicity, or disability of its proposed PDs/PIs and co-PDs/PIs. To gather information needed for this important task, the applicant should submit the requested information for each identified PD/PI and co-PDs/PIs with each proposal. Submission of the requested information is voluntary and is not a precondition of award. However, information not submitted will seriously undermine the statistical validity, and therefore the usefulness, of information received from others. Any individual not wishing to submit some or all the information should check the box provided for this purpose. Upon receipt of the application, this form will be separated from the application. This form will not be duplicated, and it will not be a part of the review process. Data will be confidential.

Co-Project Director/co-Principal Investigator 1

Prefix:	*** First Name:**	**Middle Name:**	*** Last Name:**	**Suffix:**
	Gwendolyn		Sibley	

Gender: Female

Race (check all that apply):
❍ American Indian or Alaska Native
❍ Asian
❍ Black or African American
❍ Native Hawaiian or Other Pacific Islander
❍ White
● Do Not Wish to Provide

Ethnicity:
Do Not Wish To Provide

Disability Status (check all that apply):
❍ Hearing
❍ Visual
❍ Mobility/Orthopedic Impairment
❍ Other
● None
❍ Do Not Wish to Provide

Citizenship:
US Citizen

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 R&R

3. DATE RECEIVED BY STATE	State Application Identifier

1. TYPE OF SUBMISSION

❍Pre-application ●Application ❍Changed/Corrected Application

2. DATE SUBMITTED
06/26/2026

Applicant Identifier

4. a. Federal Identifier

b. Agency Routing Identifier

c. Previous Grants.gov Tracking ID

5. APPLICANT INFORMATION **UEI:** CFMMM5JM7HJ9

Legal Name: Kansas State University
Department: Office of Sponsored Programs
Division: Office of the Vice President for Research
Street1: 1601 Vattier Street
Street2: 103 Fairchild Hall
City: Manhattan
County/Parish: Riley
State: KS: Kansas
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 66506-2504

Person to be contacted on matters involving this application

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Mr.	Paul	R.	Lowe	

Position/Title: Assoc Vice President for Research
Department:
Division:
Street1: 1601 Vattier Street
Street2: 103 Fairchild Hall
City: Manhattan
County/Parish: Riley
State: KS: Kansas
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 66506-2504
Phone Number: 785-532-6804
Fax Number: 785-532-3640
Email: plowe@K-State.edu

6. EMPLOYER IDENTIFICATION NUMBER*(EIN) or (TIN)*: 480771751

7. TYPE OF APPLICANT: H: Public/State Controlled Institution of Higher Education
Other (Specify):
Small Business Organization Type ❍Women Owned ❍Socially and Economically Disadvantaged

8. TYPE OF APPLICATION:

●New ❍Resubmission
❍Renewal ❍Continuation ❍Revision

If Revision, mark appropriate box(es).

❍A. Increase Award ❍B. Decrease Award ❍C. Increase Duration
❍D. Decrease Duration ❍E. Other*(specify)*:

Is this application being submitted to other agencies? ❍ Yes ● No What other Agencies?

9. NAME OF FEDERAL AGENCY:
National Institute of Food and Agriculture

10. ASSISTANCE LISTING NUMBER:
10.233
ASSISTANCE LISTING TITLE: Open Data Standards

11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT:
Ag-DEx: A DOI-first Open Data Framework for Agricultural Data Exchange, Discovery, and Governed Reuse

12. PROPOSED PROJECT:

Start Date	Ending Date
01/01/2027	12/31/2029

13. CONGRESSIONAL DISTRICT OF THE APPLICANT:
KS-001

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
	David		Molik	

Position/Title: Associate Professor
Organization Name: Kansas State University
Department: Hale Library
Division: K-State Libraries
Street1: 1117 MID CAMPUS DR NORTH
Street2: 113 HALE
City: Manhattan
County/Parish: Riley
State: KS: Kansas
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 66506-2504
Phone Number: 785-532-5399
Fax Number:
Email: molik@ksu.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested	$923,325.00
b. Total Non-Federal Funds	$0.00
c. Total Federal & Non-Federal Funds	$923,325.00
d. Estimated Program Income	$0.00

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?

a. YES ○ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:

DATE:

b. NO ● PROGRAM IS NOT COVERED BY E.O. 12372; OR

○ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

● I agree

The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLLL (Disclosure of Lobbying Activities) or other Explanatory Documentation. File Name: Mime Type:

19. Authorized Representative

Prefix:	First Name:	Middle Name:	Last Name:	Suffix:
Mr.	Paul	R.	Lowe	

Position/Title: Assoc Vice President for Research
Organization Name: Kansas State University
Department: Office of Sponsored Programs
Division: Vice President for Research
Street1: 1601 Vattier Street
Street2: 103 Fairchild Hall
City: Manhattan
County/Parish: Riley
State: KS: Kansas
Province:
Country: USA: UNITED STATES
ZIP / Postal Code: 66506-2504
Phone Number: 785-532-6804
Fax Number: 785-532-3640
Email: plowe@K-State.edu

Signature of Authorized Representative	Date Signed
Paul R. Lowe	06/26/2026

20. Pre-application File Name: Mime Type:

21. Cover Letter Attachment File Name: Mime Type:

Project/Performance Site Location(s)

Project/Performance Site Primary Location

Organization Name: Kansas State University

* Street1: 1601 Vattier Street		Street2: 103 Fairchild Hall
* City: Manhattan	County: Riley	* State: KS: Kansas
Province:	* Country: USA: UNITED STATES	* Zip / Postal Code: 66506-2504
UEI: CFMMM5JM7HJ9	* Project/Performance Site Congressional District: KS-001	

File Name	Mime Type

Additional Location(s)

RESEARCH & RELATED Other Project Information

1. * Are Human Subjects Involved? ❍ Yes ● No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ❍ Yes ❍ No

If yes, check appropriate exemption number

Exemption Number: _ 1 _ 2 _ 3 _ 4 _ 5 _ 6 _ 7 _ 8

If no, is the IRB review Pending? ❍ Yes ❍ No

IRB Approval Date:

Human Subject Assurance Number

2. * Are Vertebrate Animals Used? ❍ Yes ● No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ❍ Yes ❍ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. * Is proprietary/privileged information included in the application? ❍ Yes ● No

4.a. * Does the Project have an Actual or Perceived Impact – positive or negative – on the environment? ❍ Yes ● No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ❍ Yes ❍ No

4.d. If yes, please explain:

5.a. * Is the research performance site designated, or eligible to be designated, as a historic place? ❍ Yes ● No

5.b. If yes, please explain:

6.a. * Does this project involve activities outside the U.S. or partnership with International Collaborators? ❍ Yes ● No

6.b. If yes, identify countries:

6.c. Optional Explanation:

7. Project Summary/Abstract	Ag-DEx Project Summary.pdf	Mime Type: application/pdf
8. Project Narrative	Ag-DEx Project Narrative.pdf	Mime Type: application/pdf
9. Bibliography & References Cited	Ag-DEx References Cited.pdf	Mime Type: application/pdf
10. Facilities & Other Resources	Ag-DEx Facilties.pdf	Mime Type: application/pdf
11. Equipment	Ag-DEx Equipment.pdf	Mime Type: application/pdf
12. Other Attachments	LettersofSupport.pdf	Mime Type: application/pdf

PROJECT SUMMARY

Instructions:
The summary is limited to 250 words. The names and affiliated organizations of all Project Directors/Principal Investigators (PD/PI) should be listed in addition to the title of the project. The summary should be a self-contained, specific description of the activity to be undertaken and should focus on: overall project goal(s) and supporting objectives; plans to accomplish project goal(s); and relevance of the project to the goals of the program. The importance of a concise, informative Project Summary cannot be overemphasized.

Title: Ag-Dex: Agricultural Data Exchange, Discovery, Governance, And Reuse

PD: Molik, David, C	**Institution:** Kansas State University
CO-PD: Sibley, Gwen	**Institution:** Kansas State University
CO-PD: PD/PI 3 Name (Last, First, MI)	**Institution:**
CO-PD: PD/PI 4 Name (Last, First, MI)	**Institution:**
CO-PD: PD/PI 5 Name (Last, First, MI)	**Institution:**
CO-PD: PD/PI 6 Name (Last, First, MI)	**Institution:**
CO-PD: PD/PI 7 Name (Last, First, MI)	**Institution:**

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse will create a neutral, secure, and sustainable open data framework for agricultural research, Extension, and producer-centered data. Agricultural data are often scattered across spreadsheets, local databases, equipment platforms, research projects, Extension programs, and specialized tools. This fragmentation makes data difficult to preserve, document, protect, cite, share, and reuse.

Ag-DEx will be led by the Open Publishing Exchange at Kansas State University Libraries. The project will deploy a repository-backed data service that supports open data publishing where public release is appropriate, while also allowing restricted, mediated, metadata-only, or producer-sensitive records when data require protection. The framework will include repository infrastructure, metadata templates, persistent identifiers, access categories, documentation workflows, and OpenAPI-documented interfaces that allow partner systems to connect with governed data packages.

The project will be demonstrated through three applied use cases. The TAPS program will support a searchable, curated, API-accessible data hub for multi-state agricultural research and Extension data. The Ag Producer Data Exchange use case will test producer-controlled data package workflows using restricted example files. The NABC-supported, EDEN-aligned emergency-preparedness use case will test documentation and data stewardship workflows that can scale during disasters, biosecurity events, and other urgent agricultural information needs.

Ag-DEx will help agricultural data become easier to preserve, discover, cite, publish, and reuse while maintaining producer control, privacy, attribution, and project boundaries. The project will also develop governance, documentation, evaluation, and sustainability practices for long-term stewardship beyond the award period.

Project Narrative

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse

A. Introduction

Overview

Agriculture is producing more data than ever, but much of it cannot yet become useful open data. The same field or watershed may generate information through producer records, irrigation systems, sensors, imagery, equipment platforms, Extension programs, research trials, conservation initiatives, and decision-support tools, yet those data often remain scattered across spreadsheets, local databases, vendor systems, agency portals, and project-specific workflows. Fragmentation limits preservation, citation, governance, connection, and reuse across institutions, states, programs, and producer relationships. It also creates a trust problem: data that are valuable for innovation, production efficiency, water stewardship, and emergency preparedness may be difficult to share because producers and program leaders cannot easily see how the data will be documented, governed, attributed, protected, or reused.

Agricultural data exchange requires more than storage. It requires a framework for moving data through a lifecycle: from working project materials, to documented and governed data packages, to citable research and Extension outputs, to openly published data products where public release is appropriate. Producers need confidence that sensitive data will not be reused unexpectedly. Researchers need citable datasets. Extension programs need practical data hubs. Digital agriculture partners need stable interfaces, shared vocabularies, and OpenAPI-documented workflows. Future AI/ML work needs data that are structured, documented, discoverable, and computationally usable under appropriate permissions.

Ag-DEx (Agricultural Data Exchange) will develop and evaluate a neutral Open Data Framework (ODF) for producer-centered, agricultural research, and Extension data exchange. The overall objective is to create a library-led framework that supports open data publishing while preserving producer control, access rules, attribution, provenance, and project boundaries. Ag-DEx will pursue four objectives: **Objective 1: Develop the trusted repository backend**; **Objective 2: Develop the OpenAPI-documented compatibility and AI-readiness layer**; **Objective 3: Evaluate the framework through TAPS, APDE, and NABC-supported EDEN-aligned use cases**; and **Objective 4: Establish governance, stakeholder advising, documentation, evaluation, and sustainability**. The payoff is a reusable ODF model that shows how open data publishing, governed access, repository stewardship, and domain-specific interoperability can work together for agricultural communities beyond the initial pilots.

Why Ag-DEx and why OPE

The project will be led by the Open Publishing Exchange (OPE) at Kansas State University Libraries under PI David Molik. OPE will serve as the neutral institutional home for repository stewardship, metadata, citation, access control, governance, open data publishing, and long-term preservation. Agricultural collaborators will provide use-case grounding through pilot participation, support letters, data requirements, and the Ag-DEx Stakeholder Advisory Committee.

Ag-DEx is library-led because the problem is not only software deployment; it is also stewardship, governance, citation, metadata, preservation, access control, public access, and reuse. OPE can connect repository services, curation practices, persistent identifiers, open publishing, documentation, and long-term stewardship while avoiding the costs and lock-in of a purely commercial platform. K-State is an appropriate home because Ag-DEx can be built alongside Beocat research computing, institutional repository planning, New Prairie Press, Extension partnerships, water and digital-agriculture expertise, agricultural biosecurity expertise, and the PI's experience in biological data infrastructure and scholarly data publishing.

Fit with the USDA-NIFA Open Data Framework program

The USDA-NIFA ODF program calls for neutral and secure data repositories and cooperatives where producers, universities, and nonprofit entities can store and share data to foster agricultural innovation, technological progress, profitability, and production efficiency (USDA National Institute of Food and Agriculture 2026). The NOFO also requires plans for participation, governance, metadata, curation, security, privacy, accessibility, FAIR principles, cyberinfrastructure integration, user support, and sustainability beyond the award (USDA National Institute of Food and Agriculture 2026). Ag-DEx responds by combining a trusted repository backend, an agricultural API compatibility layer, governed access categories, persistent identifier-based discovery, and real-world evaluation through TAPS, the Ag Producer Data Exchange, and National Agricultural Biosecurity Center (NABC)-supported, EDEN-aligned Extension and emergency-preparedness workflows.

The compatibility layer will translate project data into reusable data packages through compatible formats, shared vocabularies, documented metadata, provenance, persistent identifiers, and access rules. This allows Ag-DEx to support open data publishing where data can be released, while also supporting metadata-only, restricted, mediated, or producer-sensitive records where open release is not appropriate.

Sustainability and infrastructure strategy

During the ODF award, TAPS, APDE, and NABC-supported EDEN-aligned pilots will be supported by project funds and will not be charged deposition or stewardship fees. The base repository will align with K-State institutional data repository efforts and support a free core of public metadata, dataset landing pages, DOI/PID citation, open documentation, discovery, and standard repository records. Beyond the award, higher-cost services (i.e., large storage, enhanced curation, restricted

access, producer-sensitive governance, API integration, and controlled computational access) may be supported through transparent fees, sponsored project budgets, service agreements, partner contributions, future grants, or cyberinfrastructure allocations.

Ag-DEx is not a monolithic platform that replaces existing agricultural systems. InvenioRDM will provide repository functions for records, files, metadata, communities, restricted access, persistent identifiers, and curation. Ag-DEx will add an agricultural compatibility and governance layer for metadata normalization, producer-sensitive access categories, APIs, discovery graphs, controlled computational access, and analytics readiness. Over 36 months, the project will deploy a high-availability InvenioRDM-based repository using two project-funded, replicated Kansas State HPC (Beocat) production nodes that expand K-State's Ceph/S3-compatible repository storage environment, plus a Terraform-defined AWS warm-standby disaster-recovery environment. Beocat will remain the repository of record; AWS will provide geographically separate recovery capacity. The project will pursue OpenAIRE integration, CoreTrustSeal alignment, and Trusted Digital Repository-readiness.

B. Rationale and Significance

Rationale

Agricultural data sharing faces two first-order barriers: access control and data compatibility. Producers, researchers, Extension professionals, and partner organizations need to share data without losing control of permissions, authorship, sensitivity, provenance, or project boundaries. At the same time, agricultural data are heterogeneous: spatial, temporal, social, biophysical, management, technology, water-use, remote sensing, precision agriculture, field-level, and public datasets may all need to be connected. Without common metadata, persistent identifiers, repository stewardship, and API-level compatibility, these data remain difficult to discover, combine, cite, publish, or reuse.

This project asks a practical research question: *How can an agricultural Open Data Framework move data toward open publishing and broad reuse while preserving producer control, governed access, and domain-specific interoperability?* Ag-DEx answers by developing and evaluating a framework that supports data through a production lifecycle: working project data, governed packages, citable research and Extension outputs, and openly published data products where release is appropriate. FAIR data can include authentication, authorization, controlled access, metadata-only records, and mediated reuse when data are sensitive or restricted (Wilkinson et al. 2016). Ag-DEx will therefore treat open data publishing as a stewardship workflow requiring metadata review, persistent identifiers, licensing or access terms, provenance, documentation, versioning, data-use conditions, and preservation commitments.

Significance

Ag-DEx is significant because it will produce and evaluate a reusable model for agricultural open data publishing that balances openness, producer trust, and computational reuse. This is important

because agriculture faces increasingly data-intensive challenges (e.g., water management, climate variability, input efficiency, conservation, emergency preparedness, animal and plant health, and producer profitability). Many of the tools needed to address these c hallenges: decision-support systems; remote-sensing pipelines; risk models; Extension dashboards; and AI/ML workflows depend on data that can be found and responsibly reused across projects. At present, agricultural data are often usable only by the project team that collected them. Ag-DEx will address that gap by turning project-specific information into governed data packages with metadata, persistent identifiers, access rules, and documentation that allow reuse while preserving trust.

Technical availability alone is not enough; reuse depends on documentation, incentives, trust, curation, infrastructure, and clarity about rights and responsibilities (Borgman 2012; Tenopir et al. 2011). Ag-DEx will pair FAIR metadata with the TRUST Principles for digital repositories (Transparency, Responsibility, User Focus, Sustainability, and Technology) (Lin et al. 2020). For Ag-DEx, this means published rules and access categories; OPE responsibility for citation, curation, provenance, and restricted access; user-centered evaluation with TAPS, APDE, and NABC-supported EDEN-aligned stakeholders; a cost-aware sustainability model; and technical infrastructure for secure, persistent, reliable services.

Ag-DEx will make agricultural data AI-ready without removing control from data owners by using structured metadata, persistent identifiers (PIDs), provenance, standardized templates, API-accessible records, and controlled computational access. It also provides a neutral third-party stewardship role: OPE is not attempting to own producer relationships, replace Extension programs, or build farm decision-support software. OPE will provide trusted infrastructure for durable storage, dataset-level citation, governance, access controls, open data publishing, and long-term stewardship.

Use cases and stakeholder support

The proposal is grounded in three use cases that allow Ag-DEx to be developed and evaluated under real conditions. First, TAPS, led by Daran Rudnick, will serve as the real-data implementation pilot. TAPS is a multi-state, Extension-connected, producer-relevant program with data-sharing challenges across institutions, including data silos, data security, authorship, cross-state reuse, and standardized data management for larger federal proposals and machine-learning applications. Ag-DEx will onboard the TAPS data backend as a searchable, curated, API-accessible data hub.

Second, APDE, associated with Dwane Roth and producer-facing collaborators, will serve as the governed producer-data workflow p ilot. APDE represents the need for a producer-controlled data exchange framework where producer-contributed data can be permissioned, described, accessed, and reused under governed conditions. The initial APDE work will focus on a generic producer data package using example files under restricted access, reducing near-term data-rights risk while developing governance patterns, access categories, metadata templates, and compatibility workflows.

Third, the NABC-supported, EDEN-aligned Extension and emergency-preparedness use case will

serve as the documentation, emergency-scaling, and scale-to-point-of-need stress test. Disaster, biosecurity, disease-response, supply-chain, and emergency-preparedness contexts create sudden demand for reliable documentation, controlled data-sharing, public-facing information products, and repeatable information-management practices. NABC will support this EDEN-aligned dimension through advisory input on agricultural biosecurity, preparedness, risk management, sensitive data, and response-oriented data stewardship.

Letters of support from TAPS, the producer/APDE perspective, Kansas Water Institute/KCARE, ID3A, and NABC reinforce the proposal's central argument. TAPS emphasizes the need for searchable, curated, API-accessible data for multi-state research and Extension collaborations. APDE emphasizes trust, transparency, permissions, and producer control. KWI/KCARE emphasizes irrigation, water conservation, remote sensing, producer engagement, and applied research. ID3A emphasizes structured, documented, permissioned data for analytics, AI/ML, remote sensing, and decision support. NABC emphasizes agricultural biosecurity, emergency preparedness, risk management, continuity planning, and safeguards for sensitive operational, geospatial, animal health, supply-chain, and emergency-response information. Together, the use cases show how Ag-DEx can preserve, cite, govern, discover, document, publish, and reuse agricultural and Extension data without collapsing ownership, authorship, permissions, or project boundaries.

C. Objectives

Ag-DEx will pursue four integrated objectives over a 36-month project period. Each objective develops and evaluates a component of a reusable agricultural Open Data Framework rather than simply deploying a local repository.

1. **Objective 1: Develop the Ag-DEx trusted repository backend.** Deploy and evaluate a high-availability InvenioRDM-based agricultural repository using two project-funded, replicated Beocat production nodes, institutional Ceph/S3-compatible object storage, backup and restore procedures, access logging, uptime monitoring, restricted access workflows, DOI/PID support, and Terraform-managed AWS warm-standby disaster recovery. Repository quality and sustainability work will include OpenAIRE integration, CoreTrustSeal alignment, and Trusted Digital Repository-readiness.
2. **Objective 2: Develop the Ag-DEx OpenAPI-documented compatibility and AI-readiness layer.** Build and evaluate an agricultural compatibility layer above the repository backend, with published OpenAPI Specification (OAS) documentation. This layer will support metadata normalization, producer-controlled access, consent-aware workflows, discovery graphs, API-based reuse, and controlled computational access for approved analytics and machine-learning workflows.
3. **Objective 3: Evaluate Ag-DEx through TAPS, APDE, and NABC-supported EDEN-aligned use cases.** Onboard the TAPS data backend as a searchable, curated, API-accessible hub

connected to Globus-enabled large-data transfer. Develop an APDE generic producer data package workflow using example files under restricted access to demonstrate producer-sensitive data exchange governance. Add a supporting NABC-supported, EDEN-aligned Extension and emergency-preparedness documentation use case to demonstrate storage of Extension, biosecurity, and emergency-preparedness documentation and to stress-test scale-to-point-of-need workflows.

4. **Objective 4: Establish governance, stakeholder advising, documentation, implementation evaluation, and sustainability.** Create published Ag-DEx rules, principles, access categories, metadata templates, security and privacy documentation, training materials, sustainability pathways, and the Ag-DEx Stakeholder Advisory Committee. The PI, Data Librarian, Research Software Engineer, Project Coordinator, and Administrative Assistant will support implementation, evaluation documentation, metadata quality, partner coordination, and sponsor reporting.

D. Approach

The approach follows the four objectives and treats implementation as the setting for framework development and evaluation. Ag-DEx will not simply deploy a repository; it will test whether a library-led Open Data Framework can move agricultural data toward open publishing, governed reuse, machine-readable discovery, and long-term stewardship while maintaining producer trust and partner-specific access controls. Each objective therefore produces both an operational deliverable and evaluation evidence: technical configuration, metadata templates, access workflows, OpenAPI documentation, pilot demonstrations, stakeholder feedback, and sustainability documentation.

Objective 1 approach: trusted repository backend

Ag-DEx will use InvenioRDM as the repository foundation. InvenioRDM will provide repository records, files, metadata, communities, restricted records/files, access requests, persistent identifiers, and curation workflows. OPE will configure and evaluate Ag-DEx as a domain-oriented agricultural data service rather than a generic file store. The repository will support open, embargoed, restricted-to-community, mediated-access, producer-sensitive/private, and metadata-only records. PIDs will be assigned through the Kansas State Libraries DataCite DOI miniting system.

The evaluation question for this objective is whether a general-purpose research data repository can be adapted into a trusted agricultural open data publishing service without losing domain-specific governance. Success will be measured not only by whether the software runs, but by whether records can be curated, cited, permissioned, transferred, backed up, restored, monitored, and documented in ways that meet agricultural stakeholder expectations. This includes testing how repository communities, metadata templates, restricted files, access requests, and persistent identifiers can be combined into repeatable agricultural workflows.

The repository backend will be designed for high availability, resilience, and documented stewardship. Two project-funded Beocat production nodes will expand K-State's Ceph/S3-compatible repository storage environment as replicated, hot, load-balanced production infrastructure supporting records,

restricted files, TAPS data, APDE examples, APIs, logging, backup/restore, monitoring, and preservation. Beocat will provide the repository of record, while AWS will provide a Terraform-defined warm-standby recovery environment using archive-grade object storage where appropriate and documented DNS failover procedures. Globus integration will connect repository records and governed data packages with large-file transfer workflows (Foster 2011). Repository quality work will include OpenAIRE integration, CoreTrustSeal alignment, and Trusted Digital Repository-readiness.

Objective 2 approach: API compatibility and AI-readiness layer

Ag-DEx will build a compatibility layer above the InvenioRDM repository API, with published OpenAPI Specification (OAS). The purpose is not to replace specialist systems or force all agricultural data into one schema. Its purpose is to provide stable, documented, governed interfaces that allow heterogeneous agricultural datasets to be deposited, described, permissioned, discovered, cited, transferred, and reused under known conditions.

The compatibility layer will expose three interface patterns. A repository orchestration interface will automate project/community creation, dataset records, metadata templates, persistent identifiers, links to projects/publications, and provenance. A domain package interface will support TAPS, APDE producer, Extension documentation, NABC-supported EDEN-aligned, and future partner packages containing files or references, metadata, access category, stewardship information, provenance, documentation, and relationships. A governed access and computation interface will expose search, metadata, transfer, and controlled computational workflows for approved users and projects. OpenAPI documentation will allow partner systems to discover endpoints, understand request and response formats, validate package structures, and integrate with governed workflows.

The evaluation question for this objective is whether compatibility can increase reuse without weakening governance. Ag-DEx will test whether access categories, stewardship roles, provenance, data-use conditions, and project/community boundaries can travel with data packages through repository records, APIs, and transfer workflows. This is essential for agriculture because a workflow that makes data technically accessible but strips away context, consent, or attribution would reduce trust rather than increase open data participation.

Ag-DEx will function as both repository and data service. A dataset can be preserved and cited, discovered through metadata and PID relationships, moved through Globus where appropriate, accessed through authorized APIs, and used by downstream analytics without removing access controls. Compatibility will not collapse governance: metadata, access categories, stewardship roles, project/community boundaries, data-use conditions, and provenance will travel with the workflow.

Objective 3 approach: TAPS, APDE, and NABC-supported EDEN-aligned use cases

The TAPS pilot will serve as the real-data implementation pilot. Ag-DEx will onboard the TAPS data backend as a searchable, curated, API-accessible data hub that functions as both a trusted repository and a dual-use data service for downstream research, Extension, decision-support, and analytics

projects. TAPS datasets will be described with agricultural metadata templates, governed through access categories, exposed through repository search and APIs, and connected to Globus-enabled transfer workflows.

The APDE pilot will serve as the governed producer-data workflow pilot. It will develop a generic producer data package workflow using example files under restricted access. This workflow will test producer-sensitive access categories, metadata-only discovery, restricted files, mediated access, provenance, and data-owner control without defaulting to open release or prematurely exposing sensitive producer data.

The NABC-supported, EDEN-aligned Extension and emergency-preparedness use case will demonstrate storage of Extension-oriented, biosecurity, and emergency-preparedness data and usage documentation while preparing for sudden scale. It will focus on preservation, metadata, discovery, reuse guidance, controlled access, and repeatable deposit workflows rather than a third full data hub. The use case will stress-test workflows that can scale to the point of need during disasters, disease events, supply-chain disruptions, or urgent Extension responses.

Objective 4 approach: governance, advising, workforce, and sustainability

Ag-DEx will be OPE-led and OPE-managed, with advisory governance through the Ag-DEx Stakeholder Advisory Committee. The committee will advise on producer needs, TAPS requirements, APDE workflows, access categories, metadata priorities, governance principles, pilot evaluation, emergency-preparedness considerations, and sustainability. Initial advisory participation will be anchored by TAPS, ID3A, Kansas Water Institute/KCARE, NABC-supported EDEN-aligned stakeholders, and the APDE producer perspective.

Ag-DEx will operate under published rules, principles, and values, including producer control, transparency, no-surprise reuse, dataset citation, stewardship, FAIR metadata, privacy, security, interoperability, sustainability, non-competition with producer-facing tools, and separation of discovery from access. The Research Software Engineer is the primary technical implementation role and is budgeted in place of a trainee position because the work requires production repository engineering, API development, infrastructure automation, and integration with research computing. The Data Librarian will support metadata, curation, citation, documentation, policy alignment, and FAIR/TRUST-aligned stewardship. The Project Coordinator will support partner workflows, advisory activities, implementation metrics, evaluation documentation, and coordination. The Administrative Assistant will support grant reporting, meeting logistics, travel coordination, documentation management, purchasing, and compliance records.

Ag-DEx's sustainability model will combine institutional repository alignment with cost-aware expansion. The base service will be sustained through K-State repository efforts; enhanced services may be sustained through future project funding, partner agreements, service arrangements, sponsored stewardship, or cyberinfrastructure allocations. Dissemination will be supported through

OPE and New Prairie Press, K-State's library-based diamond open-access publishing program. No publication charges are requested.

Access categories and governance controls

Ag-DEx will support multiple access categories so that different data types can be governed appropriately. The categories are intentionally broader than a simple open/closed distinction.

Access category	Purpose
Open	Public records and files available for reuse under stated terms.
Embargoed	Records or files released after a defined period.
Restricted to project/community	Records or files visible or accessible only to authorized groups.
Mediated access	Data available only by request and approval under stated conditions.
Producer-sensitive/private	Data governed by producer or project permissions and not openly released.
Metadata-only	Discovery record available without public file access.

These categories will be implemented through repository permissions, community/project roles, restricted files, mediated access workflows, metadata templates, and documented stewardship policies. The access model will allow Ag-DEx to make data discoverable and citable while preserving control over sensitive files and attributes.

Potential problems and contingency plans

If TAPS onboarding requires more harmonization than expected, the team will prioritize a documented subset and use remaining data types to refine templates. If APDE data-rights concerns arise, the project will use synthetic or representative restricted files while preserving the governance workflow. If AWS recovery costs or requirements change, Beocat will remain the repository of record and the project will document a scaled recovery model. If stakeholders find an access category or metadata template unclear, the team will revise and retest it with advisory input.

E. Project Timeline

Ag-DEx will be implemented over 36 months, allowing time for repository deployment, Beocat production configuration, AWS warm-standby disaster recovery, API development, TAPS onboarding, APDE restricted workflows, Globus integration, governance documentation, training, evaluation, and sustainability planning.

Period	Major activities	Key deliverables
Months 1-6	Hire project staff; convene Ag-DEx Stakeholder Advisory Committee; confirm infrastructure architecture; begin InvenioRDM configuration; begin TAPS backend assessment; define APDE generic producer package; identify NABC-supported EDEN-aligned documentation deposit requirements.	Staffing complete; advisory committee launched; architecture plan; draft access categories; NABC-supported EDEN-aligned deposit requirements identified.
Months 7-12	Deploy repository backend; configure Beocat production nodes; configure backup/restore, logging, monitoring; begin metadata templates; begin Globus integration; begin OpenAIRE/CoreTrustSeal/TDR-readiness documentation; begin AWS warm-standby design and Terraform templates.	Working repository prototype; Beocat production nodes configured; draft metadata templates; Globus integration plan; AWS warm-standby design; quality documentation outline.
Months 13-18	Onboard TAPS backend; build API compatibility layer; implement search and API access; test restricted access workflows; develop APDE restricted example package; begin NABC-supported emergency-preparedness documentation onboarding; implement AWS replication procedures.	TAPS hub beta; API beta; APDE package prototype; access workflow test; NABC-supported EDEN-aligned collection beta; replication workflow test.

Period	Major activities	Key deliverables
Months 19-24	Expand TAPS hub; refine metadata normalization; develop DOI/PID discovery graph; test controlled computational access; test warm-standby recovery and DNS failover procedures; conduct stakeholder review.	TAPS hub v1; discovery graph prototype; controlled computation test; documented recovery test; stakeholder feedback report.
Months 25-30	Finalize APDE workflow; complete training materials; strengthen OpenAIRE/CoreTrustSeal alignment; refine sustainability model; evaluate pilots using implementation metrics and advisory feedback; finalize NABC-supported EDEN-aligned documentation workflow.	APDE workflow v1; training materials; repository quality documentation; sustainability draft; Extension documentation workflow completed.
Months 31-36	Final implementation evaluation; finalize documentation; publish rules/principles/values; complete sustainability plan; prepare future funding/allocation requests; disseminate outputs through OPE/New Prairie Press as appropriate.	Final repository service; final TAPS hub; final APDE workflow; final NABC-supported EDEN-aligned documentation collection; final sustainability plan; final report.

F. Project Management Plan

Ag-DEx will be led by David Molik, Department Head of OPE at K-State Libraries. OPE will serve as the managing unit and neutral institutional home for repository, metadata, citation, access-control, governance, open data publishing, and long-term stewardship. The project will not use a large co-PI structure; instead, it will combine centralized OPE leadership, grant-funded implementation

staff, and structured collaboration through pilot partners and the Ag-DEx Stakeholder Advisory Committee.

The PI will oversee project direction, repository strategy, partner coordination, infrastructure design, quality planning, grant reporting, and alignment with Beocat, Research IT, K-State Libraries, OPE, New Prairie Press, and agricultural partners. The Research Software Engineer will lead InvenioRDM deployment/configuration, high-availability Beocat integration, Ag-DEx API compatibility development, Terraform-managed AWS warm-standby failover, Globus integration, OpenAIRE interoperability, monitoring/logging, and technical documentation. The Data Librarian will support metadata, curation, dataset citation, documentation, repository policy alignment, training materials, and FAIR/TRUST-aligned stewardship workflows.

The Project Coordinator will coordinate partner workflows, advisory committee activities, implementation metrics, evaluation documentation, pilot workflow tracking, and collaboration across TAPS, APDE, and NABC-supported EDEN-aligned activities. The Administrative Assistant will support grant reporting, meeting logistics, travel coordination, documentation management, purchasing, sponsor compliance records, and directly allocable project administration.

The Ag-DEx Stakeholder Advisory Committee will include representation or structured input from TAPS, ID3A, Kansas Water Institute/KCARE, the APDE producer perspective, and NABC-supported EDEN-aligned emergency-preparedness and agricultural biosecurity stakeholders. Support letters document stakeholder demand and applicability rather than creating a co-PI or subaward structure. TAPS documents the need for a multi-state data hub and repository backend. Dwane Roth/APDE documents producer-centered data exchange needs, trust, transparency, permissions, and producer control. ID3A documents durable repository infrastructure, metadata, persistent identifiers, access controls, API compatibility, governance, and long-term stewardship for digital agriculture and AI/ML-ready analytics. Kansas Water Institute/KCARE documents irrigation, water, conservation, remote sensing, producer engagement, and applied research needs. NABC documents governed agricultural data discovery, biosecurity preparedness, risk management, and controlled access for sensitive operational, geospatial, animal health, supply-chain, and emergency-response information.

Evaluation and success measures

Ag-DEx will use implementation evaluation to assess whether the project produces a functional, trusted, stakeholder-informed agricultural data framework. Evaluation will focus on five questions: whether the repository backend is operational and reliable; whether TAPS, APDE, and NABC-supported EDEN-aligned use cases can be supported through repeatable workflows; whether metadata and APIs make data discoverable and computationally usable under governed conditions; whether stakeholders find governance and access categories understandable and useful; and whether the sustainability model is credible beyond the award period.

The evaluation is intentionally practical. The project will not measure success only by system launch,

but by evidence that the framework can be used, understood, and extended. Advisory feedback, workflow demonstrations, metadata reviews, API tests, transfer tests, access-control checks, and documentation reviews will be used to identify which parts of the framework are ready for continued service, which require revision, and which should be generalized for future agricultural data projects.

Evaluation area	Measures of progress and success
Repository readiness	InvenioRDM deployed; Beocat production nodes configured; backup/restore tested; access logging and monitoring operational; AWS warm-standby replication and disaster-recovery plan documented.
Trust alignment	TRUST principles mapped to policies; CoreTrustSeal self-assessment initiated; OpenAIRE pathway documented; TDR-readiness documentation drafted.
TAPS pilot	TAPS backend onboarding plan completed; curated records created; metadata templates applied; API access tested; Globus workflow demonstrated.
APDE pilot	Generic producer data package workflow created; restricted example files deposited; permission/access workflow tested; producer-facing review completed.
NABC-supported EDEN-aligned use case	Documentation collection deposited; usage documentation created; emergency-scaling workflow documented; repeatable templates produced.
Governance and users	Ag-DEx Stakeholder Advisory Committee convened; access categories reviewed; feedback documented; policies revised based on advisory input.
AI readiness and sustainability	Machine-readable metadata records created; API discovery tested; controlled computational access pattern demonstrated; free-core and cost-recovery model documented and reviewed.

Evaluation evidence will include technical checklists, repository configuration documentation, metadata completeness reviews, API tests, Globus transfer tests, warm-standby disaster-recovery tests, pilot workflow demonstrations, advisory committee notes, training/documentation outputs, and a final sustainability review. The Project Coordinator will support evaluation by documenting pilot workflows, tracking implementation metrics, synthesizing advisory feedback, and preparing evaluation summaries for annual and final reporting. If stakeholder surveys or interviews are later added, the project will seek appropriate K-State IRB review or exemption before collecting human-subjects data.

G. Data Management Plan

Ag-DEx will generate repository configuration documentation, metadata templates, API documentation, governance materials, stakeholder feedback summaries, training materials, pilot dataset records, TAPS repository metadata, APDE generic producer package metadata, restricted example files, and NABC-supported EDEN-aligned Extension and emergency-preparedness data and usage documentation records. It will also generate evaluation data related to repository performance, access workflows, metadata completeness, API usability, implementation milestones, documentation use, and stakeholder feedback.

Project-generated documentation, metadata templates, governance materials, training materials, and non-sensitive evaluation outputs will be deposited in Ag-DEx or another appropriate K-State repository with persistent identifiers and open licenses where appropriate. These outputs are part of the project's open data publishing contribution: they will allow other agricultural programs to see how Ag-DEx structures access categories, metadata requirements, data package templates, workflow documentation, and sustainability planning. Software or configuration code will be shared through a public code repository when security and institutional policy allow. Sensitive configuration details, security documentation, access logs, restricted files, producer-sensitive data, and private stakeholder information will not be openly released.

Pilot data will be managed according to Ag-DEx access categories: open, embargoed, restricted to project/community, mediated access, producer-sensitive/private, and metadata-only. TAPS data will be curated into searchable records with metadata, persistent identifiers where appropriate, API access, and Globus-enabled transfer workflows; access levels will be determined with TAPS leadership and data stewards. APDE materials will use a generic producer package workflow with example files under restricted access, with metadata-only and mediated access where appropriate. NABC-supported EDEN-aligned materials will be managed as Extension, emergency-preparedness, and agricultural biosecurity documentation and data records, emphasizing practical documentation, controlled access, rapid onboarding, and scale-to-point-of-need workflows.

Each deposited data package will include a minimum metadata profile sufficient for discovery, citation, governance, and reuse. Required elements will include title, creator or steward, organizational

affiliation, project or community, abstract or description, temporal and spatial coverage where applicable, data type, package contents, provenance, access category, license or access terms, recommended citation, related publications or projects, funding acknowledgement, version, and contact or mediated-access instructions. Domain-specific extensions will be developed for TAPS, APDE, and NABC-supported EDEN-aligned materials where needed, including management variables, sensor or equipment context, geospatial references, documentation type, emergency-preparedness context, data-use limitations, and sensitivity or risk considerations.

Each pilot workflow will include curation review before open publication, restricted release, or metadata-only publication. The Data Librarian will review metadata completeness, citation readiness, documentation, access-category language, and reuse terms. The Research Software Engineer will support technical validation of package structure, API exposure, storage location, backup/restore status, and Globus transfer documentation where applicable. The PI or delegated OPE steward will approve access-category assignments for records involving restricted, mediated, producer-sensitive, or security-sensitive materials. This review process is intended to move data toward open publication when appropriate while ensuring that restricted records still have reliable discovery, provenance, and mediated-access pathways.

Ag-DEx will preserve versioned records so that changes to data packages, documentation, metadata, access conditions, and reuse terms can be tracked over time. Open records will receive persistent identifiers, including DataCite DOIs, where appropriate. Restricted and metadata-only records may also receive persistent landing pages when this supports discovery, citation, or mediated access without exposing sensitive files. Storage, retention, preservation, and access commitments will be documented at the record or collection level and managed according to institutional capacity, partner agreements, data-owner permissions, access category, and stewardship requirements.

Ag-DEx metadata will support FAIR principles in a rights-aware manner by making datasets findable, accessible under stated conditions, interoperable through structured metadata and APIs, and reusable through provenance, documentation, citation, and governance conditions. The PI will oversee the DMP, with OPE responsible for repository stewardship, metadata workflows, access categories, and preservation planning. The Data Librarian will support metadata quality, curation, citation, and documentation; the Research Software Engineer will support API documentation, storage workflows, backup/restore, AWS warm-standby procedures, and repository integration; and the Project Coordinator will support implementation documentation, pilot tracking, evaluation summaries, and advisory feedback documentation. Long-term stewardship will be supported through OPE and K-State institutional repository efforts, with project documentation, metadata templates, Extension usage documentation, and non-sensitive governance materials remaining available after the award period.

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Facilities and Other Resources

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse

Institutional Environment

Ag-DEx will be housed at Kansas State University Libraries in the Open Publishing Exchange (OPE). OPE provides an institutional setting that combines open publishing, repository stewardship, metadata development, persistent identifier w orkflows, pu blic ac cess su pport, documentation practices, and long-term preservation planning. This environment is well matched to Ag-DEx because the project is not only a software deployment; it is an open data publishing and stewardship framework for agricultural data that must support curation, citation, governance, access controls, documentation, and reuse over time.

Kansas State University provides a strong institutional environment for agricultural data infrastructure. The project will draw on K-State Libraries, OPE, New Prairie Press, K-REx, Beocat research computing, Ceph/S3-compatible storage planning, Extension partnerships, water and digital-agriculture programs, and agricultural collaborators. The co-location of library-based publishing, repository services, research computing, and agricultural expertise creates a practical environment for developing and evaluating a neutral, secure, and sustainable Open Data Framework for producer-centered, research, Extension, and emergency-preparedness data.

OPE will serve as the operational home for Ag-DEx. The unit will provide project leadership, repository stewardship, metadata and citation practices, open data publishing guidance, documentation workflows, and coordination with K-State Libraries and project p artners. OPE's library-based position is important because Ag-DEx requires a neutral backend that can support producer-facing and Extension-facing programs without competing with those programs for producer relationships, decision-support services, or commercial value. By locating the framework in OPE, Ag-DEx will provide horizontal data infrastructure: a shared stewardship, repository, metadata, citation, access-control, and governance layer that supports multiple agricultural programs while allowing each program to maintain its own relationships, tools, audiences, and data ownership.

Beocat Research Computing, Storage, and Data Transfer

Beocat is Kansas State University's primary research computing environment and provides the institutional cyberinfrastructure setting for Ag-DEx. Beocat supports data-intensive research across the university and provides the operational context needed for production repository services. Beocat supports OpenStack infrastructure which will be utilized by Ag-DEx. The availability of Beocat is central to Ag-DEx because the framework must support agricultural data packages that are larger, more heterogeneous, and more operationally sensitive than ordinary document deposits.

Ag-DEx funds will support two high-storage-capacity Beocat production nodes that expand K-State's repository and research computing environment for the proposed agricultural data framework.

These nodes will provide production capacity for InvenioRDM repository services, restricted files, access-controlled agricultural data packages, API-accessible services, logging, monitoring, backup and restore workflows, and preservation o perations. The nodes will be integrated with K-State's Ceph/S3-compatible storage environment and research computing strategy rather than operated as short-term project servers. This approach gives Ag-DEx a production-oriented home inside existing K-State cyberinfrastructure. Beocat serverspace itself is access controlled through campus rfid ID systems.

Beocat's role is also important for data movement and partner workflows. Agricultural datasets may include field-level files, sensor data, imagery, management records, documentation packages, and derived analytical outputs. The Beocat environment provides a practical setting for connecting repository records to data-transfer and computational workflows. K-State already has a Beocat Globus endpoint, and Ag-DEx will use Globus-enabled workflows to support efficient transfer of larger data packages between repository, computational, and partner environments. This capacity is especially important for TAPS and future agricultural data hubs that require reliable large-file movement without relying on ad hoc transfer methods.

Ag-DEx will also use a cost-controlled hybrid disaster-recovery environment. Beocat will remain the production repository of record, while Amazon Web Services will provide geographically separate warm-standby recovery capacity. AWS resources will support replicated recovery data where appropriate, archive-grade object storage, DNS or failover support, repository service testing, and related technical services. This model provides resilience and recovery capacity while keeping the durable production service anchored in K-State infrastructure.

Repository and Documentation Infrastructure: K-REx and Ag-DEx

Kansas State University Libraries and the Open Publishing Exchange provide repository and documentation infrastructure that will support the project. K-REx, the K-State Research Exchange, provides an institutional repository environment for preserving and disseminating scholarly and institutional outputs. For Ag-DEx, K-REx provides an existing institutional context for public documentation, reports, repository policy documents, and other non-sensitive project outputs that should remain accessible beyond the grant period (i.e., Grey Literature depoisiton). K-REx runs on DSpace 10 and assigns Crossref DOIs to deposted digital materials.

Ag-DEx will complement this institutional repository environment by developing a domain-oriented agricultural data service. Ag-DEx will use InvenioRDM as its repository foundation for records, files, metadata, communities, restricted access, and curation workflows. Where K-REx is appropriate for public-facing documentation and institutional outputs, Ag-DEx will provide the governed agricultural data layer needed for data packages, access categories, metadata-only discovery, producer-sensitive records, mediated access, and reusable data products.

This repository environment will allow the project to connect datasets, documentation, metadata,

publications, workflows, and persistent identifiers. Public project documentation, user guides, access-category documentation, training materials, and final reports can be published through K-REx or other appropriate K-State repository channels. Agricultural data packages and related metadata will be managed through Ag-DEx according to their access category. Together, K-REx and Ag-DEx provide complementary infrastructure for open documentation, governed data stewardship, citation, preservation, and long-term discovery.

Open Publishing and Dissemination Infrastructure

New Prairie Press is K-State Libraries' diamond open-access publishing program and provides an important dissemination pathway for Ag-DEx. New Prairie Press supports library-led publishing for journals, conference proceedings, monographs, and other scholarly and public-interest outputs. For Ag-DEx, New Prairie Press and OPE provide an institutional publishing environment that can support open data publishing guidance, public documentation, and scholarly communication outputs without relying on commercial article processing charges. No article processing charges are requested in the Ag-DEx budget.

Project outputs that are well suited to New Prairie Press may include reports, technical documentation, Extension-facing materials, and articles or reports submitted to the *Journal of the Kansas Agricultural Experiment Station*. This venue is a natural fit for Kansas agricultural and Extension-facing outputs that document project results, use cases, workflows, and practical implications for applied agricultural audiences.

Ag-DEx dissemination will not be limited to New Prairie Press. The project may also submit scholarly outputs to appropriate diamond open access external non-NPP venues such as the *Journal of Librarianship and Scholarly Communication* and the *Journal of Extension*. These venues are relevant to the project's library-led infrastructure, open data publishing, scholarly communication, Extension, and applied outreach dimensions. The availability of both K-State library publishing infrastructure and appropriate external journals gives Ag-DEx multiple dissemination routes for technical, scholarly, and practitioner audiences.

This open publishing environment strengthens the project as Ag-DEx is explicitly concerned with the movement of project data and documentation toward publishable, citable, and reusable outputs. The same institutional setting that supports repository records and data stewardship also supports open documentation, reports, journal outputs, and persistent connections between datasets, publications, and workflows.

Agricultural, Extension, and Emergency-Preparedness Partner Environment

Ag-DEx will be developed and evaluated in an agricultural environment with active partner input. TAPS will provide the real-data implementation use case and will help ground the framework in multi-state agricultural research and Extension data needs. The Ag Producer Data Exchange perspective will provide producer-centered requirements around permissions, producer control,

restricted access, metadata-only discovery, and mediated reuse. Kansas Water Institute/KCARE and ID3A provide water, digital agriculture, analytics, and producer-relevant perspectives that will help ensure the framework is useful beyond a single program or dataset type.

The National Agricultural Biosecurity Center (NABC) will support the EDEN-aligned emergency-preparedness dimension of the project. NABC's involvement strengthens the project's capacity to address agricultural biosecurity, risk management, disease response, continuity planning, sensitive data, controlled access, and response-oriented data stewardship. This environment is important for testing whether Ag-DEx workflows can scale to the point of need during disasters, biosecurity events, supply-chain disruptions, or other urgent agricultural information needs.

These partner environments provide data requirements, stakeholder feedback, advisory input, and practical use cases rather than subaward-funded implementation. The technical and stewardship work will be carried out by OPE, the Research Software Engineer, the Data Librarian, the Project Coordinator, and the Administrative Assistant, while partners will help evaluate whether Ag-DEx is usable, trustworthy, and aligned with agricultural, producer, Extension, and emergency-preparedness needs.

Personnel, Workspace, and Administrative Support

Project personnel will be based in the K-State Libraries and OPE environment, with access to library workspace, meeting space, collaboration tools, and administrative support. The PI will provide overall project leadership, repository strategy, partner coordination, infrastructure planning, governance, sustainability planning, and reporting oversight. The Data Librarian will support metadata design, curation workflows, dataset citation, documentation, repository policy alignment, training materials, and FAIR/TRUST-aligned stewardship practices. OPE offices are within Hale Library at Kansas State Univeristy Main Campus.

The Research Software Engineer will have access to the K-State Libraries, OPE, and Beocat environments needed to support repository deployment and integration. The Project Coordinator and Administrative Assistant will have access to the workspace, meeting support, documentation systems, and administrative resources needed to coordinate partner engagement, advisory committee activities, reporting, purchasing, and sponsor compliance. K-State Libraries and OPE provide a collaborative setting for coordinating technical implementation, data stewardship, open publishing, partner engagement, and grant administration.

Capability Statement

Together, K-State Libraries, OPE, Beocat, Ceph/S3-compatible storage, Globus-enabled transfer, K-REx, New Prairie Press, and the project's agricultural and emergency-preparedness partners provide a strong facilities environment for Ag-DEx. These resources create a production-oriented setting for developing and evaluating an Open Data Framework that can preserve, govern, cite, publish, document, discover, transfer, and reuse agricultural data while protecting producer-sensitive

and restricted information. The institutional environment is sufficient to support the proposed 36-month project and to provide a realistic pathway for sustaining the base repository service after the award period.

Equipment

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse

Equipment

Ag-DEx requests $65,000 in Budget Period 1 for high-capacity Beocat production nodes. These nodes will expand K-State's institutional repository and research computing environment to support production Ag-DEx repository operations.

The equipment is necessary because Ag-DEx will operate as a repository-backed Open Data Framework, not simply a development prototype. The nodes will support InvenioRDM repository services, restricted files, access-controlled agricultural data packages, metadata and persistent identifier workflows, OpenAPI-documented services, logging, monitoring, backup/restore workflows, and preservation operations.

The equipment will be integrated with K-State's Beocat research computing environment and Ceph/S3-compatible repository storage strategy. Beocat will serve as the production repository environment for Ag-DEx, while separate AWS services will provide warm-standby disaster recovery. The requested equipment will support high-availability local production capacity for the repository of record.

Existing K-State resources provide the institutional computing, storage, and support environment, but dedicated production nodes are needed to ensure reliable repository performance, controlled access, and long-term stewardship for the TAPS, APDE, and NABC-supported EDEN-aligned use cases. The equipment will remain part of K-State's research cyberinfrastructure and will support the sustainability of the base Ag-DEx repository service after the award period.

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June 22, 2026

Dear Open Data Framework Review Committee Members:

On behalf of Kansas State University's National Agricultural Biosecurity Center (NABC), I am pleased to provide this letter of support for the Ag-DEx proposal led by Dr. David Molik and the Open Publishing Exchange at Kansas State University Libraries.

The NABC works with government agencies, academic institutions, agricultural organizations, and private-sector partners to strengthen the preparedness and resilience of the nation's food and agricultural systems. Across this work, access to reliable, well-documented, and appropriately governed data is essential for risk assessment, emergency preparedness, disease response, continuity planning, and informed decision-making. However, agricultural data are frequently distributed across separate programs, institutions, technologies, and information systems, limiting their discoverability, interoperability, and long-term value.

Ag-DEx addresses this challenge by proposing a neutral and trusted framework for agricultural data discovery, governance, and responsible reuse. Its planned capabilities—including durable repository infrastructure, metadata management, persistent identifiers, governed access controls, application programming interface compatibility, and long-term stewardship—would help ensure that agricultural data remain understandable, accessible, and useful across projects, organizations, and time.

From an agricultural biosecurity perspective, effective data exchange must balance accessibility with appropriate safeguards. Agricultural datasets may contain producer-sensitive, operational, geospatial, animal health, supply-chain, or other information that could create privacy, cybersecurity, economic, or biosecurity risks if accessed or reused without sufficient context and controls. Ag-DEx's emphasis on clearly defined permissions, responsible governance, and controlled access is therefore critical to building the trust required for meaningful participation by producers, researchers, government agencies, and industry partners.

The proposed framework also has the potential to strengthen preparedness and response capabilities by making relevant data easier to locate, integrate, and responsibly apply. Improved data stewardship can support analytical tools, modeling, situational awareness, after-action learning, and interdisciplinary research addressing animal and plant health threats, natural disasters, supply-chain disruptions, and other hazards affecting agriculture.

The NABC expects to support the project through advisory participation, feedback concerning agricultural biosecurity and risk-management considerations, and connections to relevant researchers, agencies, and partner networks. Ag-DEx would strengthen Kansas State University's capacity to support secure, collaborative, and data-informed agricultural research and engagement while contributing to the resilience of food and agricultural systems nationally. The National Agricultural Biosecurity Center strongly supports this proposal and encourages USDA-NIFA's favorable consideration of this important effort.

Sincerely,

Marty Vanier DVM

Marty Vanier, DVM
Director, National Agricultural Biosecurity Center
mvanier@bri.ksu.edu
www.k-state.edu/nabc

June 17, 2026

Subject: Letter of Support for Ag-DEx: Agricultural Data Exchange: Discovery, Governance, and Reuse

Dear Review Committee Members,

I am pleased to provide this letter of support for the Ag-DEx proposal led by Dr. David Molik and the Open Publishing Exchange at Kansas State University Libraries.

The Testing Ag Performance Solutions (TAPS) program brings together producers, researchers, Extension specialists, and industry partners to evaluate agricultural management decisions under real-world production conditions. Through this work, TAPS generates diverse datasets related to agronomic management, environmental performance, economic outcomes, technology evaluation, decision-support research, and crop response. These data support research, Extension activities, and producer-centered learning across multiple states and production environments.

As TAPS continues to grow, the volume and diversity of data generated through the program create increasing opportunities and challenges for long-term stewardship, documentation, discovery, governance, and reuse across years, institutions, and collaborating states. Ag-DEx addresses these needs by providing a trusted framework for agricultural data management and exchange. The proposed infrastructure would support repository services, metadata management, persistent identifiers, governed access controls, and long-term stewardship. These capabilities can help improve the discoverability, interoperability, and usability of agricultural data while supporting responsible data sharing and future analytical applications.

From the TAPS perspective, Ag-DEx offers an opportunity to create a searchable, curated, and API-accessible data environment that can support multi-state research and Extension collaborations and increase the long-term value of producer-centered datasets. The project aligns closely with TAPS interests in advancing agricultural innovation, Extension engagement, and data-informed decision-making.

The TAPS program strongly supports this proposal and appreciates USDA-NIFA's consideration of this important effort.

Sincerely,



Daran R. Rudnick, Ph.D.
Professor and Director of Sustainable Irrigation
Carl and Melinda Helwig Department of Biological and Agricultural Engineering
Kansas State University
1059 Seaton Hall
920 N. Martin Luther King, Jr. Drive
Manhattan, KS 66506
Email: drudnick@ksu.edu
Office: 712-204-6772

June 15, 2026

USDA-NIFA Open Data Framework Review Committee

Re: Letter of Support for Ag-DEx: Agricultural Data Exchange: Discovery and Governed Reuse

Dear Review Committee Members:

My name is Dwane Roth. I am a fourth-generation farmer from Holcomb, Kansas, and I have spent many years working with producers, researchers, Extension specialists, conservation organizations, and industry partners on practical approaches to agricultural water management and agricultural innovation. I am writing in support of the Ag-DEx proposal led by Dr. David Molik and the Open Publishing Exchange at Kansas State University.

Over the past several years, I have been working on the concept of an Agricultural Producer Data Exchange designed to help producers better manage, organize, and securely share agricultural data. Farmers generate large amounts of information through equipment, irrigation systems, sensors, yield maps, field records, conservation programs, and research projects. Unfortunately, much of that information remains isolated in separate systems, making it difficult to integrate, analyze, or use effectively.

The challenge is not simply collecting data. The challenge is to create a trusted framework that enables data to be discovered, connected, and reused while maintaining producer control. Farmers need confidence that they understand who has access to their information, how it will be used, and whether it will be shared beyond its intended purpose. Without trust and transparency, participation will always be limited.

That is why I support the Ag-DEx proposal. The project recognizes that agricultural data sharing requires more than technology. It requires governance, clear permissions, metadata standards, and secure methods for managing access. Producer data should not automatically become public simply because it is useful. Instead, data exchanges should provide a neutral and trusted environment where ownership, permissions, and use rights remain clearly defined.

I believe this type of infrastructure is increasingly important as agriculture becomes more data-driven. Producers are being asked to participate in research projects, conservation programs, sustainability initiatives, water management efforts, and emerging market opportunities that all require access to reliable information. Too often, producers are required to provide the same information to multiple organizations. Better data frameworks can reduce that burden while improving the value of the information being collected.

I also see a direct connection to the TAPS (Testing Ag Performance Solutions) program. I have participated in the Kansas TAPS program for the past two years and have seen firsthand how valuable it can be when producers, researchers, and Extension work together around real-world management decisions. Programs such as TAPS generate important datasets that can advance agricultural innovation, but they also require effective systems to organize, document, share, and govern data across farms, years, institutions, and states.

The Agricultural Producer Data Exchange concept is ultimately about making data more useful without taking control away from producers. Ag-DEx provides many of the foundational elements needed to support that vision, including data discovery, metadata management, governance frameworks, access controls, and interoperability across systems. These capabilities can help researchers accelerate innovation while ensuring that producers remain trusted partners in the process.

I support this proposal because it addresses a real and growing need within agriculture. As our industry continues to generate more information, we need systems that make data easier to use, share responsibly, and trust. Ag-DEx represents an important step toward building that future.

Sincerely,

Dwane Roth

Dwane Roth
Front Porch Farms LLC

June 15, 2026

USDA-NIFA Open Data Framework Review Committee

Re: Letter of Support for Ag-DEx: Agricultural Data Exchange: Discovery, and Governed Reuse

Dear Review Committee Members:

On behalf of the Board of Directors of Kansas State University's Institute for Digital Agriculture and Advanced Analytics (ID3A), we are pleased to provide this letter of support for the Ag-DEx proposal led by Dr. David Molik and the Open Publishing Exchange at Kansas State University Libraries.

ID3A supports interdisciplinary work that advances digital agriculture, data science, analytics, and practical decision-support tools for agriculture. A persistent challenge in this work is that agricultural data are often fragmented across research projects, technologies, equipment platforms, sensors, imagery tools, and institutional repositories. These data have significant potential value, but that value is limited when datasets are not well described, governed, discoverable, or usable across systems.

Ag-DEx addresses this foundational challenge. The proposed framework would provide a neutral backend for agricultural data exchange, including durable repository infrastructure, metadata, persistent identifiers, access controls, API compatibility, governance, and long-term stewardship. These capabilities are central to the future of digital agriculture. Advanced analytics, artificial intelligence, machine learning, remote sensing, and decision-support tools all depend on data that are structured, documented, permissioned, and accessible under appropriate conditions.

ID3A expects to support the project through advisory participation, feedback on digital agriculture and analytics needs, and connection to relevant K-State researchers, programs, and partner networks. Ag-DEx would strengthen K-State's capacity to support data-intensive agricultural research and applied innovation. The ID3A Board of Directors strongly supports this proposal and encourages USDA-NIFA's favorable consideration.

Sincerely,

Trevor Hefley

On behalf of the Board of Directors
Institute for Digital Agriculture and Advanced Analytics (ID3A)
Kansas State University

KANSAS STATE UNIVERSITY

June 15, 2026

USDA-NIFA Open Data Framework Review Committee

Re: Letter of Support for Ag-DEx: Agricultural Data Exchange: Discovery, and Governed Reuse

Dear Review Committee Members:

I am pleased to provide this letter of support for the Ag-DEx proposal led by Dr. David Molik and the Open Publishing Exchange at Kansas State University Libraries. The Kansas Water Institute (KWI) strongly supports the development of a neutral, well-governed agricultural data framework that can serve producers, researchers, Extension professionals, and partner organizations.

Kansas water challenges require practical, producer-relevant data systems. Across our work in irrigation, water conservation, remote sensing, producer engagement, and applied research, we regularly see the same challenge: useful data are generated across many programs and platforms, but they are often difficult to preserve, describe, connect, govern, and reuse. This limits the value of public and private investment in agricultural and water research. It also makes it harder to develop tools that are useful to producers and decision-makers.

Ag-DEx addresses a real need by creating a trusted repository and long-term stewardship. The proposal also recognizes that open data in agriculture cannot mean uncontrolled release of producer-sensitive information. Producers need confidence that their data will not be reused without clear permission, context, and safeguards. Ag-DEx's emphasis on governed access and producer control is essential for building the trust needed to support meaningful data sharing.

The project aligns closely with KWI's mission to support water research, education, and engagement that help Kansas address current and future water resource challenges. By strengthening the infrastructure needed to preserve and responsibly reuse agricultural and water data, Ag-DEx will improve K-State's ability to support interdisciplinary research and practical solutions for Kansas and the region.

I strongly support this proposal and appreciate USDA-NIFA's consideration.

Sincerely,



Susan Metzger

Director, Strategic Interdisciplinary Program Development
Director, Kansas Water Institute (KWI)
Director, Institute for Digital Agriculture and Advanced Analytics (ID3A) – Engagement
Director, Kansas Center for Agricultural Resources and the Environment (KCARE)
Office of the Vice President for Research

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator

Prefix: First Name*: David Middle Name Last Name*: Molik Suffix:

Position/Title*:	Associate Professor
Organization Name*:	Kansas State University
Department:	Hale Library
Division:	K-State Libraries
Street1*:	1117 MID CAMPUS DR NORTH
Street2:	113 HALE
City*:	Manhattan
County:	Riley
State*:	KS: Kansas
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	66506-2504

Phone Number*: 785-532-5399 Fax Number: E-Mail*: molik@ksu.edu

Credential, e.g., agency login:

Project Role*: PD/PI Other Project Role Category:

Degree Type: PhD Degree Year: 2021

	File Name
Attach Biographical Sketch*:	cv-2840043-3Molik.pdf
Attach Current & Pending Support:	cpos-2827554Molik.pdf

PROFILE - Senior/Key Person

Prefix: First Name*: Gwendolyn Middle Name Last Name*: Sibley Suffix:

Position/Title*:	Assistant Professor
Organization Name*:	Kansas State University
Department:	Hale Library
Division:	K-State Libraries
Street1*:	113 Hale Library
Street2:	1117 Mid-Campus Drive North
City*:	Manhattan
County:	Riley
State*:	KS: Kansas
Province:	
Country*:	USA: UNITED STATES
Zip / Postal Code*:	66506-1103

Phone Number*: 7855323514 Fax Number: E-Mail*: grsibley@ksu.edu

Credential, e.g., agency login:

Project Role*: Co-PD/PI Other Project Role Category:

Degree Type: Degree Year:

	File Name
Attach Biographical Sketch*:	cv-2841679-sibley.pdf
Attach Current & Pending Support:	cpos-2848314-USDA.pdf

IDENTIFYING INFORMATION:

NAME: Molik, David

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-3192-6538

POSITION TITLE: Department Head, Associate Professor, Librarian

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
USDA ARS TPGMBRU, Hilo, Hawaii, United States	Postdoctoral Fellow	06/2021 - 03/2023	Computational Biology
University of Notre Dame, Notre Dame, Indiana, United States	PHD	08/2016 - 05/2021	Integrated BioMedical Sciences
Rensselaer Polytechnic Institute, Troy, New York, United States	BS	08/2009 - 12/2012	Information Technology and Web Science

Appointments and Positions

2025 - present Department Head, Associate Professor, Librarian, Kansas State University, Manhattan, Kansas, United States

2023 - 2025 Computational Biologist - SY , USDA ARS ABADRU , Manhattan, Kansas, United States

2021 - 2023 Postdoctoral Fellow, USDA ARS TPGMBRU, Hilo, Hawaii, United States

2020 - 2021 Graduate Scholar, Navari Center for Digital Scholarship, University of Notre Dame, Notre Dame, Indiana, United States

2016 - 2021 Graduate Research Appointments (Included research appointments with the Emrich Lab, Genomics and Bioinformatics Core, and Goodson Lab, with work in bioinformatics, scientific computing, and data-intensive biology), University of Notre Dame, South Bend, Indiana, United States

2014 - 2016 Scientific Informatics Developer, Cold Spring Harbor Laboratory, Cold Spring Harbor, New York, United States

2013 - 2014 Software Engineer / Network Operations Center Manager, Synacor Inc, Buffalo, New York, United States

2012 - 2012 NSF REU, Corso Lab, SUNY Buffalo, Computer Science and Engineering, Amherst, New York, United States

2010 - 2013 Undergraduate Research and Internship Appointments, Rensselaer Polytechnic Institute, Troy, New York, United States

Products

1. Cabugos Leyla, Buble Katheryn, Daenzer Jenna, Jung Sook, Main Dorrie, Marrano Annarita,

Molik David, Raciti Daniela, Wright Adam, Yook Karen, Reiser Leonore. AgBioDatabase Finder: an online tool to help researchers find and submit agricultural genomic, genetic, and breeding data. 2025. Available from: https://www.micropublication.org/journals/biology/micropub-biology-001896 DOI: 10.17912/MICROPUB.BIOLOGY.001896

2. Cannon E, Molik D, Wright A, Zhang H, Honaas L, Chougule K, Dyer S. Guidelines for gene and genome assembly nomenclature. GENETICS. 2025 March; 229(3):-. Available from: https://academic.oup.com/genetics/article/doi/10.1093/genetics/iyaf006/7954581 DOI: 10.1093/genetics/iyaf006
3. Adam Wright, Mark D. Wilkinson, Christopher Mungall, Scott Cain, Stephen Richards, Paul W. Sternberg, Ellen Provin, Jonathan Jacobs, Scott Geib, Daniela Raciti, Karen J. Yook, Lincoln Stein, David Molik. FAIR Header Reference genome: a TRUSTworthy standard. Briefings in Bioinformatics. 2024 March; 25(3):bbae122. Available from: https://doi.org/10.1093/bib/bbae122 DOI: 10.5281/zenodo.6762550
4. Adam Wright, Mark D Wilkinson, Christopher Mungall, Scott Cain, Stephen Richards, Paul Sternberg, Ellen Provin, Jonathan L Jacobs, Scott Geib, Daniela Raciti, Karen Yook, Lincoln Stein, David C Molik. FAIR Header Reference genome: a TRUSTworthy standard. Briefings in Bioinformatics. 2024 March. DOI: 10.5281/zenodo.6762550
5. Molik DC. met v1: expanding on old estimations of biodiversity from eDNA with a new database framework. Database (Oxford). 2022 May 11;2022 PubMed Central PMCID: PMC9216496.
6. Walker K, Kalra D, Lowdon R, Chen G, Molik D, Soto DC, Dabbaghie F, Khleifat AA, Mahmoud M, Paulin LF, Raza MS, Pfeifer SP, Agustinho DP, Aliyev E, Avdeyev P, Barrozo ER, Behera S, Billingsley K, Chong LC, Choubey D, De Coster W, Fu Y, Gener AR, Hefferon T, Henke DM, Höps W, Illarionova A, Jochum MD, Jose M, Kesharwani RK, Kolora SRR, Kubica J, Lakra P, Lattimer D, Liew CS, Lo BW, Lo C, Lötter A, Majidian S, Mendem SK, Mondal R, Ohmiya H, Parvin N, Peralta C, Poon CL, Prabhakaran R, Saitou M, Sammi A, Sanio P, Sapoval N, Syed N, Treangen T, Wang G, Xu T, Yang J, Zhang S, Zhou W, Sedlazeck FJ, Busby B. The third international hackathon for applying insights into large-scale genomic composition to use cases in a wide range of organisms. F1000Res. 2022;11:530. PubMed Central PMCID: PMC9557141.
7. Molik D.C., Tomlinson D., Davitt S., Morgan E.L., Sisk M., Roche B., Meyers N., Pfrender M.E.. Combining natural language processing and metabarcoding to reveal pathogenenvironment associations. PLoS Neglected Tropical Diseases. 2021; 15(4). DOI: 10.1371/journal.pntd.0008755
8. Molik D, Pfrender M, Emrich S. Effects from structure of Metabarcode Sequences on Lossy Analysis of Microbiome Data. 2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). 2018 IEEE International Conference on Bioinformatics and Biomedicine (BIBM); ; Madrid, Spain. IEEE; c2018. Available from: https://ieeexplore.ieee.org/document/8621143/ DOI: 10.1109/BIBM.2018.8621143
9. Molik D, Pfrender M, Emrich S. Uncovering Effects from the Structure of Metabarcode Sequences for Metagenetic and Microbiome Analysis. Methods and Protocols. 2020 March 12;

3(1):22-. Available from: https://www.mdpi.com/2409-9279/3/1/22 DOI: 10.3390/mps3010022

10. Ong J, Pence J, Molik D, Shepherd H, Goodson H. Yeast grown in continuous culture systems can detect mutagens with improved sensitivity relative to the Ames test. PLOS ONE. 2021; 16(3):e0235303-. Available from: https://dx.plos.org/10.1371/journal.pone.0235303 DOI: 10.1371/journal.pone.0235303
11. Molik DC, Tomlinson D, Davitt S, Morgan EL, Sisk M, Roche B, Meyers N, Pfrender ME. Combining natural language processing and metabarcoding to reveal pathogen-environment associations. PLoS Negl Trop Dis. 2021 Apr;15(4):e0008755. PubMed Central PMCID: PMC8055023.
12. Molik DC. An Outsider's Perspective on Why We Climb Mountains and Why Projects Like the i5k Matter. J Insect Sci. 2022 Jul 1;22(4) PubMed Central PMCID: PMC9250708.
13. Scroggs S, Bird E, Molik D, Nayduch D. Vesicular Stomatitis Virus Elicits Early Transcriptome Response in Culicoides sonorensis Cells. Viruses. 2023 October 18; 15(10):2108-. Available from: https://www.mdpi.com/1999-4915/15/10/2108 DOI: 10.3390/v15102108
14. Molik D, Stahlke A, Sharma S, Simmonds T, Corpuz R, Kauwe A, Schrader J, Mason C, Sim S, Geib S. otb: an automated HiC/HiFi pipeline assembles the *Prosapia bicincta* Genome. G3: Genes, Genomes, Genetics. 2024 August; 14(8):-. Available from: https://academic.oup.com/g3journal/article/doi/10.1093/g3journal/jkae129/7691216 DOI: 10.1093/g3journal/jkae129
15. Drolet B, Reister-Hendricks L, Mayo C, Rodgers C, Molik D, McVey D. Increased Virulence of Culicoides Midge Cell-Derived Bluetongue Virus in IFNAR Mice. Viruses. 2024 September 17; 16(9):1474-. Available from: https://www.mdpi.com/1999-4915/16/9/1474 DOI: 10.3390/v16091474
16. David C Molik, Amanda R Stahlke, Sharu P Sharma, Tyler J Simmonds, Renee L Corpuz, Angela N Kauwe, Jeremy E Schrader, Charles J Mason, Sheina B Sim, Scott M Geib. otb: an automated HiC/HiFi pipeline assembles the Prosapia bicincta Genome. G3: Genes, Genomes, Genetics. 2024 August. DOI: 10.1093/g3journal/jkae129

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Molik, David in SciENcv on 2026-06-25 09:49:55

IDENTIFYING INFORMATION:

NAME: Sibley, Gwendolyn

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-6552-9472

POSITION TITLE: Scholarly Communication and Data Librarian

PRIMARY ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Emporia State University, Emporia, Kansas, United States	MLS	08/2020 - 05/2022	Librarianship
University of Kansas, Lawrence, Kansas, United States	AB	08/2016 - 05/2020	English Rhetoric, Literature, & Writing

Appointments and Positions

2025 - present Scholarly Communication and Data Librarian, Kansas State University, Manhattan, Kansas, United States

2022 - 2025 Scholarly Communication and Copyright Librarian, Kansas State University, Manhattan, Kansas, United States

Products

1. Turvey-Welch, M., Sibley, G., Otto, R., Kickbusch, S., Jackson, C., Berney, E. Enjoying “First Sale Doctrine”: An Appreciation of library lending from then to now. New Prairie Press; 2025. Available from: https://newprairiepress.org/cgi/viewcontent.cgi?article=1106&context=culsproceedings PMID: https://doi.org/10.4148/ 2160-942X.1106
2. Hall, D., Sibley. G. Welcome to the Legal Circus. US: Kansas Library Association; 2024.
3. Sibley, G. Indigenous Intellectual Property: Growing the Conversation. Association of Tribal Archives, Libraries, & Museums; 2024.
4. Sibley, G. The Incongruous Relationship Between AI Development & University Regulatory Policy. Kraemer Copyright Conference; 2024.
5. Spahr, K., McCowen, R., Jackson, C., Otto, R., Sibley, G., Berney, E., Turvey-Welch, M., Misilo, T., Andresen, D., Molik, D. Empowering Data Literacy. K-State Libraries Inaugural Internal Grant; 2024.
6. Jackson, C., Sibley, G., & Davis, M. Open For All. Open Education Talks; 2024.
7. Sibley, G. Copyright and AI: Do you Agree with the Terms and Conditions?. US: Library of Congress FEDLINK Spring Expo; 2023.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Sibley, Gwendolyn in SciENcv on 2026-06-26 12:05:58

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Molik, David

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-3192-6538

*POSITION TITLE: Department Head, Associate Professor, Librarian

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	RCN: Reimagining a Sustainable Data Network to Accelerate Agricultural Research and Discovery
***Status of Support:**	Current
Proposal/Award Number:	2126334-005
***Source of Support:**	Phoenix Bioinformatics Corp, Prime: NSF
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2026
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$95,729

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.75

***Overall Objectives:**

Support coordination, standards development, and community infrastructure for agricultural genomics databases through the AgBioData Research Coordination Network, including activities related to metadata practices, data integration, and community engagement.

***Statement of Potential Overlap:**

No scientific or budgetary overlap. The AgBioData RCN supports community coordination, standards development, and network-building activities for agricultural genomics databases. In contrast, the proposed project focuses on the development and empirical evaluation of machine-actionable publication frameworks and research data infrastructure. There is no duplication of research objectives or committed effort.

***Proposal/Active Project Title:**	Ag-DEx: A DOI-first Open Data Framework for Agricultural Data Exchange, Discovery, and Governed Reuse
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USDA NIFA
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$923,325

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.29
2028	0.29
2029	0.29

***Overall Objectives:**

Ag-DEx will create a neutral, secure, and sustainable open data framework for agricultural research, Extension, and producer-centered data. The project will deploy an InvenioRDM-based repository backend, persistent identifier workflows, metadata templates, access categories, OpenAPI-documented compatibility interfaces, and cyberinfrastructure connections for storage, transfer, backup, and recovery. Ag-DEx will demonstrate and evaluate the framework through three applied use cases: a TAPS data hub for multi-state agricultural research and Extension data; an Ag Producer Data Exchange workflow for producer-controlled data packages; and an Extension/Natural Disasters and NABC-supported EDEN-aligned documentation workflow for emergency-scaling and practical Extension use. The project will also establish governance rules, stakeholder advising, and a sustainability model to make agricultural data easier to discover, cite, publish, and reuse while maintaining producer control and project boundaries.

***Statement of Potential Overlap:**

This is the proposed USDA-NIFA Open Data Framework project. There is no overlap to report within this entry beyond the effort and budget requested for this proposal.

***Proposal/Active Project Title:**	Unlocking Agricultural Research Data for Better Farming Decisions
***Status of Support:**	Pending

Proposal/Award Number:

***Source of Support:** Foundation for Food & Agriculture Research (FFAR)

***Primary Place of Performance:** Kansas State University, Manhattan, KS, USA

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2029

***Total Anticipated Proposal/Project Amount:** $449,912

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

Develop an innovative research program in food and agricultural systems focused on advancing data-driven and computational approaches to improve biological understanding, data integration, and decision-making in agricultural contexts.

***Statement of Potential Overlap:**

No scientific or budgetary overlap. The FFAR nomination represents a broad early-career research program in food and agriculture focused on data-driven and computational approaches to improve biological understanding, data integration, and decision-making. The proposed USDA-NIFA Ag-DEx project focuses specifically on developing and evaluating a neutral open data framework, repository infrastructure, governance model, and applied agricultural data exchange workflows. There is no duplication of specific aims, methods, budget, or committed effort. David Molik will manage effort across projects within institutional and sponsor-approved commitments.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Molik, David in SciENcv on 2026-06-25 09:55:03

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Sibley, Gwendolyn

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-6552-9472

*POSITION TITLE: Scholarly Communication and Data Librarian

*ORGANIZATION AND LOCATION: Kansas State University, Manhattan, Kansas, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Ag-DEx: A DOI-first Open Data Framework for Agricultural Data Exchange, Discovery, and Governed Reuse
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USDA NIFA
***Primary Place of Performance:**	Kansas State University, Manhattan, KS, USA
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2027
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$923,325

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.31
2028	0.31
2029	0.31

***Overall Objectives:**

Ag-DEx will create a neutral, secure, and sustainable open data framework for agricultural research, Extension, and producer-centered data. The project will deploy an InvenioRDM-based repository backend, persistent identifier workflows, metadata templates, access categories, OpenAPI documented compatibility interfaces, and cyberinfrastructure connections for storage, transfer, backup, and recovery. Ag-DEx will demonstrate and evaluate the framework through three applied use cases: a TAPS data hub for multi-state agricultural research and Extension data; an Ag Producer Data Exchange workflow for producer-controlled data packages; and an Extension/Natural Disasters and NABC-supported EDEN-aligned documentation workflow for emergency-scaling and practical Extension use. The project will also establish governance rules, stakeholder advising, and a sustainability model to make agricultural data easier to discover, cite, publish, and reuse while maintaining producer control and project boundaries.

***Statement of Potential Overlap:**

This is the proposed USDA-NIFA Open Data Framework project. There is no overlap to report within this entry beyond the effort and budget requested for this proposal.
The Data Librarian will support metadata, curation, dataset citation, documentation, repository policy alignment, training materials, and FAIR/TRUST-aligned stewardship workflows.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Sibley, Gwendolyn in SciENcv on 2026-06-26 10:38:24

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2027 *** End Date:** 12-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary ($)	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits ($)	* Funds Requested ($)
1.	David		Molik		PD/PI	92,700.00	0.29			2,248.00	764.00	3,012.00
2.	Gwendolyn		Sibley		Co PI	66,950.00	0.31			1,718.00	584.00	2,302.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: Mime Type: **Total Senior/Key Person** **5,314.00**

B. Other Personnel

* Number of Personnel	* Project Role	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits	* Funds Requested ($)
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
1	Secretarial/Clerical	12			15,000.00	6,900.00	21,900.00
1	Research Software Engineer	12			90,000.00	30,600.00	120,600.00
1	Project Coordinator	12			21,667.00	7,367.00	29,034.00
3	**Total Number Other Personnel**					**Total Other Personnel**	**171,534.00**
					Total Salary, Wages and Fringe Benefits (A+B)		**176,848.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2027 *** End Date:** 12-31-2027 **Budget Period: 1**

C. Equipment Description

List items and dollar amount for each item exceeding $5,000

Equipment Item	* Funds Requested ($)
1. 2 high-capacity Beocat production nodes	65,000.00

Total funds requested for all equipment listed in the attached file

Total Equipment **65,000.00**

Additional Equipment: File Name: Mime Type:

D. Travel	Funds Requested ($)
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	
2. Foreign Travel Costs	
Total Travel Cost	

E. Participant/Trainee Support Costs	Funds Requested ($)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	**Total Participant/Trainee Support Costs**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 1

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2027 *** End Date:** 12-31-2027 **Budget Period: 1**

F. Other Direct Costs	**Funds Requested ($)**
1. Materials and Supplies	5,000.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	5,339.00
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	**10,339.00**

G. Direct Costs	**Funds Requested ($)**
Total Direct Costs (A thru F)	**252,187.00**

H. Indirect Costs

Indirect Cost Type	**Indirect Cost Rate (%)**	**Indirect Cost Base ($)**	*** Funds Requested ($)**
1. On Campus AGR 45.5% MTDC	45.5	187,187.00	85,170.00
		Total Indirect Costs	**85,170.00**

Cognizant Federal Agency US Department of Health and Human Services, Olulola Oluborode, 214-767-3261
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	**Funds Requested ($)**
Total Direct and Indirect Institutional Costs (G + H)	**337,357.00**

J. Fee	**Funds Requested ($)**

K. Total Costs and Fee	**Funds Requested ($)**
Total Costs and Fee (I + J)	**337,357.00**

L. * Budget Justification	File Name: Ag-DEx Budget Justification.pdf	Mime Type: application/pdf
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 2

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2028 *** End Date:** 12-31-2028 **Budget Period: 2**

A. Senior/Key Person

Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary ($)	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits ($)	* Funds Requested ($)
1.	David		Molik		PD/PI	95,481.00	0.29			2,315.00	787.00	3,102.00
2.	Gwendolyn		Sibley		Co PI	68,958.00	0.31			1,770.00	602.00	2,372.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: Mime Type: **Total Senior/Key Person** **5,474.00**

B. Other Personnel

* Number of Personnel	* Project Role	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits	* Funds Requested ($)
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
1	Secretarial/Clerical	12			15,450.00	7,108.00	22,558.00
1	Research Software Eng	12			92,700.00	31,518.00	124,218.00
1	Project Coordinator	12			22,316.00	7,588.00	29,904.00
3	**Total Number Other Personnel**					**Total Other Personnel**	**176,680.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**182,154.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 2

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2028 *** End Date:** 12-31-2028 **Budget Period: 2**

C. Equipment Description

List items and dollar amount for each item exceeding $5,000

Equipment Item	* Funds Requested ($)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment: File Name: Mime Type:

D. Travel	Funds Requested ($)
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	1,887.00
2. Foreign Travel Costs	
Total Travel Cost	**1,887.00**

E. Participant/Trainee Support Costs	Funds Requested ($)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	

Number of Participants/Trainees **Total Participant/Trainee Support Costs**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 2

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2028 *** End Date:** 12-31-2028 **Budget Period: 2**

F. Other Direct Costs	**Funds Requested ($)**
1. Materials and Supplies	
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	11,260.00
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	**11,260.00**

G. Direct Costs	**Funds Requested ($)**
Total Direct Costs (A thru F)	**195,301.00**

H. Indirect Costs

Indirect Cost Type	**Indirect Cost Rate (%)**	**Indirect Cost Base ($)**	*** Funds Requested ($)**
1. On Campus AGR 45.5% MTDC	45.5	195,301.00	88,862.00
		Total Indirect Costs	**88,862.00**

Cognizant Federal Agency US Department of Health and Human Services, Olulola Oluborode, 214-767-3261
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	**Funds Requested ($)**
Total Direct and Indirect Institutional Costs (G + H)	**284,163.00**

J. Fee	**Funds Requested ($)**

K. Total Costs and Fee	**Funds Requested ($)**
Total Costs and Fee (I + J)	**284,163.00**

L. * Budget Justification	File Name: Ag-DEx Budget Justification.pdf	Mime Type: application/pdf
	(Only attach one file.)	

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 3

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2029 *** End Date:** 12-31-2029 **Budget Period: 3**

A. Senior/Key Person

Prefix	* First Name	Middle Name	* Last Name	Suffix	* Project Role	Base Salary ($)	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits ($)	* Funds Requested ($)
1.	David		Molik		PD/PI	98,300.00	0.29			2,384.00	811.00	3,195.00
2.	Gwendolyn		Sibley		Co PI	71,027.00	0.31			1,823.00	620.00	2,443.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: Mime Type: **Total Senior/Key Person** **5,638.00**

B. Other Personnel

* Number of Personnel	* Project Role	Cal. Months	Acad. Months	Sum. Months	* Requested Salary ($)	* Fringe Benefits	* Funds Requested ($)
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
1	Secretarial/Clerical	12			15,914.00	7,320.00	23,234.00
1	Research Software Eng	12			95,481.00	32,464.00	127,945.00
1	Project Coordinator	12			22,986.00	7,815.00	30,801.00
3	**Total Number Other Personnel**					**Total Other Personnel**	**181,980.00**
					Total Salary, Wages and Fringe Benefits (A+B)		**187,618.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 3

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2029 *** End Date:** 12-31-2029 **Budget Period: 3**

C. Equipment Description

List items and dollar amount for each item exceeding $5,000

Equipment Item	* Funds Requested ($)

Total funds requested for all equipment listed in the attached file

Total Equipment

Additional Equipment: File Name: Mime Type:

D. Travel	Funds Requested ($)
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	1,992.00
2. Foreign Travel Costs	
Total Travel Cost	**1,992.00**

E. Participant/Trainee Support Costs	Funds Requested ($)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	

Number of Participants/Trainees **Total Participant/Trainee Support Costs**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-L, BUDGET PERIOD 3

*** UEI:** CFMMM5JM7HJ9

*** Budget Type:** ● Project ❍ Subaward/Consortium

Enter name of Organization: Kansas State University

*** Start Date:** 01-01-2029 *** End Date:** 12-31-2029 **Budget Period: 3**

F. Other Direct Costs	**Funds Requested ($)**
1. Materials and Supplies	
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	17,816.00
5. Subawards/Consortium/Contractual Costs	
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
Total Other Direct Costs	**17,816.00**

G. Direct Costs	**Funds Requested ($)**
Total Direct Costs (A thru F)	**207,426.00**

H. Indirect Costs

Indirect Cost Type	**Indirect Cost Rate (%)**	**Indirect Cost Base ($)**	*** Funds Requested ($)**
1. On Campus AGR 45.5% MTDC	45.5	207,426.00	94,379.00
		Total Indirect Costs	**94,379.00**

Cognizant Federal Agency US Department of Health and Human Services, Olulola Oluborode, 214-767-3261
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	**Funds Requested ($)**
Total Direct and Indirect Institutional Costs (G + H)	**301,805.00**

J. Fee	**Funds Requested ($)**

K. Total Costs and Fee	**Funds Requested ($)**
Total Costs and Fee (I + J)	**301,805.00**

L. * Budget Justification File Name: Ag-DEx Budget Justification.pdf Mime Type: application/pdf
(Only attach one file.)

RESEARCH & RELATED Budget {F-L} (Funds Requested)

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals ($)	
Section A, Senior/Key Person		**16,426.00**
Section B, Other Personnel		**530,194.00**
Total Number Other Personnel	9	
Total Salary, Wages and Fringe Benefits (A+B)		**546,620.00**
Section C, Equipment		**65,000.00**
Section D, Travel		**3,879.00**
1. Domestic	3,879.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		**39,415.00**
1. Materials and Supplies	5,000.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services	34,415.00	
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1		
9. Other 2		
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		**654,914.00**
Section H, Indirect Costs		**268,411.00**
Section I, Total Direct and Indirect Costs (G + H)		**923,325.00**
Section J, Fee		
Section K, Total Costs and Fee (I + J)		**923,325.00**

Budget Justification

Ag-DEx: Agricultural Data Exchange, Discovery, Governance, and Reuse

A. Senior/Key Personnel

Funds are requested for senior/key personnel effort necessary to provide scientific, technical, and project leadership. Salary requests are based on institutional salary records and include standard institutional escalation in later budget periods. Fringe benefits are calculated according to Kansas State University rates used in the submitted budget.

David Molik, PD/PI. Funds are requested for 0.29 calendar months in each budget period. The PI will provide overall project leadership; repository and open data publishing strategy; coordination with OPE, K-State Libraries, Beocat, Research IT, TAPS, APDE, Extension, ID3A, and KWI/KCARE partners; oversight of governance, sustainability, reporting, and budget execution; and final responsibility for project deliverables. Requested salary/fringe totals are $3,012 in Budget Period 1, $3,102 in Budget Period 2, and $3,195 in Budget Period 3, for a cumulative total of $9,309.

Gwendolyn Sibley, Data Librarian, Senior/Key Personnel. Funds are requested for 0.31 calendar months in each budget period. The Data Librarian will support metadata design, curation workflows, dataset citation, open data publishing guidance, repository policy alignment, training materials, and FAIR/TRUST-aligned stewardship practices. This effort is essential to transforming agricultural project data into documented, citable, governed, and publishable data products. Requested salary/fringe totals are $2,302 in Budget Period 1, $2,372 in Budget Period 2, and $2,443 in Budget Period 3, for a cumulative total of $7,117.

Total senior/key personnel funds requested: $16,426.

B. Other Personnel

Other personnel funds support the implementation workforce required to build, operate, document, coordinate, and administer the Ag-DEx cyberinfrastructure project. The project does not request graduate student or postdoctoral support; instead, funds are concentrated on technical implementation and project operations.

Research Software Engineer. Funds are requested for 12 calendar months in each budget period. The Research Software Engineer will serve as the primary technical implementation role for Ag-DEx. Responsibilities will include InvenioRDM deployment and configuration; development of the OpenAPI-documented Ag-DEx compatibility layer; Beocat/Ceph/S3 repository integration; Globus workflow integration; Terraform-managed AWS warm-standby disaster recovery configuration; monitoring, logging, backup/restore, API documentation, and technical handoff materials. Requested salary/fringe totals are $120,600 in Budget Period 1, $124,218 in Budget Period 2, and $127,945 in Budget Period 3, for a cumulative total of $372,763.

Project Coordinator. Funds are requested for 12 calendar months in each budget period at a

part-time project level. The Project Coordinator will support partner workflows, advisory committee coordination, implementation metrics, evaluation documentation, pilot workflow tracking, annual and final reporting support, and coordination across TAPS, APDE, Extension/Natural Disasters, and EDEN-aligned activities. Requested salary/fringe totals are $29,034 in Budget Period 1, $29,904 in Budget Period 2, and $30,801 in Budget Period 3, for a cumulative total of $89,739.

Administrative Assistant / Secretarial-Clerical Support. Funds are requested for 12 calendar months in each budget period at a part-time project level. The Administrative Assistant will provide directly allocable project support for meeting logistics, scheduling, travel coordination, purchasing, documentation management, support-letter and advisory-committee records, sponsor compliance records, and grant reporting logistics. These duties are necessary because Ag-DEx is a sole-PI, OPE-managed infrastructure project involving multiple partners, advisory activities, purchases, services, and reporting requirements. Requested salary/fringe totals are $21,900 in Budget Period 1, $22,558 in Budget Period 2, and $23,234 in Budget Period 3, for a cumulative total of $67,692.

Total other personnel funds requested: $530,194. Total salary, wages, and fringe benefits requested for all personnel: $546,620.

C. Equipment

Funds are requested in Budget Period 1 for high-capacity Beocat production nodes ($65,000). These nodes will expand K-State's institutional repository and research computing environment to support Ag-DEx production repository operations. The equipment will support InvenioRDM repository services, restricted files, access-controlled agricultural data packages, API-accessible services, logging, backup/restore workflows, monitoring, and preservation operations. The equipment is necessary for high-availability production repository capacity and for aligning Ag-DEx with K-State's Beocat and Ceph/S3-compatible storage environment.

Total equipment funds requested: $65,000.

D. Travel

Domestic travel funds are requested in Budget Period 2 ($1,887) and Budget Period 3 ($1,992), for a cumulative total of $3,879. Travel will support project coordination, stakeholder engagement, advisory and partner meetings, training or dissemination activities, and direct engagement with agricultural, Extension, or repository/cyberinfrastructure partners as needed. No foreign travel is requested.

Total travel funds requested: $3,879.

E. Participant/Trainee Support Costs

No participant/trainee support costs are requested.

F. Other Direct Costs

Materials and Supplies. Funds are requested in Budget Period 1 for materials and supplies ($5,000). These funds will support directly allocable project implementation needs such as documentation and training supplies, minor hardware or peripherals not meeting the equipment threshold, repository implementation materials, data curation support materials, and other supplies necessary to launch and document the Ag-DEx service.

ADP/Computer Services. Funds are requested for ADP/computer services in Budget Period 1 ($5,339), Budget Period 2 ($11,260), and Budget Period 3 ($17,816), for a cumulative total of $34,415. These costs will support cloud and computing services necessary for the Ag-DEx hybrid cyberinfrastructure model, including AWS warm-standby disaster recovery resources, archive-grade object storage where appropriate, DNS/failover support, repository service testing, monitoring, and related technical services needed to support reliable repository operations and recovery workflows. Beocat will remain the production repository of record; these services provide cost-controlled geographically separate recovery capacity and operational support.

Publication Costs / Article Processing Charges (APCs). $0 is requested for publication costs or APCs. Project dissemination and open data publishing outputs will use OPE, K-State Libraries, and New Prairie Press library-based publishing infrastructure where appropriate. New Prairie Press does not require article processing charges for these project dissemination pathways, allowing Ag-DEx documentation, reports, and related public-interest outputs to be shared without commercial publishing fees.

Total other direct costs requested: $39,415.

G. Total Direct Costs

Total direct costs requested are $654,914 across the three-year project period. Budget Period 1 direct costs are $252,187; Budget Period 2 direct costs are $195,301; and Budget Period 3 direct costs are $207,426.

H. Indirect Costs

Indirect costs are calculated using Kansas State University's on-campus Agricultural Research rate of 45.5% applied to the Modified Total Direct Cost base. Equipment is excluded from the MTDC base. The MTDC base is $187,187 in Budget Period 1, $195,301 in Budget Period 2, and $207,426 in Budget Period 3. Indirect costs are $85,170 in Budget Period 1, $88,862 in Budget Period 2, and $94,379 in Budget Period 3, for a cumulative total of $268,411.

I. Total Project Costs

Total direct and indirect costs requested are $923,325. Budget Period 1 total costs are $337,357; Budget Period 2 total costs are $284,163; and Budget Period 3 total costs are $301,805. No fee is requested.

Budget Summary

Budget Category	BP1	BP2	BP3 / Cumulative
Senior/Key Personnel	$5,314	$5,474	$5,638 / $16,426
Other Personnel	$171,534	$176,680	$181,980 / $530,194
Total Salary, Wages, and Fringe	$176,848	$182,154	$187,618 / $546,620
Equipment	$65,000	$0	$0 / $65,000
Travel	$0	$1,887	$1,992 / $3,879
Participant/Trainee Support	$0	$0	$0 / $0
Publication Costs / APCs	$0	$0	$0 / $0
Other Direct Costs	$10,339	$11,260	$17,816 / $39,415
Total Direct Costs	$252,187	$195,301	$207,426 / $654,914
Indirect Costs	$85,170	$88,862	$94,379 / $268,411
Total Costs	$337,357	$284,163	$301,805 / $923,325

Supplemental Information Form

Please complete this form in conjunction with the SF-424 Application for Federal Financial Assistance.

1. Funding Opportunity

* Funding Opportunity Name

Open Data Framework

* Funding Opportunity Number

USDA-NIFA-OTHER-011816

2. Program to which you are applying

* Program Code Name

Open Data Framework

* Program Code

ODF

*** 3. Type of Applicant**

H: Public/State Controlled Institution of Higher Education

4. Additional Applicant Types

5. Supplemental Applicant Types *(Check all that apply)*

- ☐ Alaska Native-Serving Institution
- ☑ Cooperative Extension Service
- ☐ Hispanic-Serving Institution
- ☐ Historically Black College or University (other than 1890)
- ☐ Minority-Serving Institution
- ☐ Native Hawaiian-Serving Institution
- ☐ Public Nonprofit Junior or Community College
- ☐ Public Secondary School
- ☐ School of Forestry
- ☑ State Agricultural Experiment Station
- ☐ Tribal College (other than 1994)
- ☑ Veterinary School or College

6. ASAP Recipient Information

* Does the legal applicant have an active Automated Standard Application for Payments (ASAP) Recipient Identification Number for NIFA awards?

● Yes ○ No

* What is the ASAP Recipient ID (which corresponds with this applications's DUNS and EIN) to be used in the event of an award?

12402200

*** 7. Key Words**

Cyber infrastructure, FAIR, TRUST,Open Data, Agricultural Data

8. Conflict of Interest List

Ag-DEx Conflict of Interest.pdf

Tracking Number:

Attachments

ConflictofInterestList

File Name	Mime Type
Ag-DEx Conflict of Interest.pdf	application/pdf

Tracking Number:

CONFLICT OF INTEREST LIST

Name: David Molik

Instructions:

- Who completes this template: Each project director/principal investigator (PD/PI) or other person that the Request for Applications (RFA) specifies
- How this template is completed:
  - List alphabetically – with last name first -- the full names of the following individuals:
    - All co-authors on publications within the past three years, including pending publications and submissions
    - All collaborators on projects within the past three years, including current and planned collaborations
    - All thesis or postdoctoral *advisees/advisors*
    - All persons in your field with whom you have had a consulting/financial arrangement/other conflict-of-interest in the past three years
  - Indicate the person's relationship to you (Co-Author, Collaborator, etc) with an "x".

Additional pages may be used as necessary.

Note: Other individuals working in the applicant's specific area are not in conflict of interest with the applicant unless those individuals fall within one of the listed categories.

Name	Co-Author	Collaborator	Advisees/ Advisors	Other – Specify Nature
Andorf, Carson		x		
Archie, Elizabeth			x	
Beal, Jake				Editorial relationship: IEEE cyberbio26 Program Committee
Bird, Edward J.	x			
Boston, Tony	x			
Bowman-Zatzkin, Whitney				Editorial relationship: Cyberbiosecurity Quarterly
Brown, Sue		x		
Buble, Katheryn	x			
Cabugos, Leyla	x			
Cain, Scott	x			
Campbell, Jaqueline	x			
Cannon, Ethalinda K. S.	x			
Carbonara, Joaquin	x			
Chougule, Kapeel	x			
Coates, Brad		x		
Coggins, Bret				Personal
Corpuz, Renee L.	x			
Daenzer, Jenna	x			
Diop, Massamba	x			
Douglas, Gavin				Editorial relationship: PCI Genomics
Drucker, Debora	x			
Dyer, Sarah	x	x		
Emrich, Scott			x	

Fry, Matt	x			
Geib, Scott M.	x			
Grellet, Sylvain	x			
Hackett, Kevin		x		
Halfon, Marc	x	x		
Honaas, Loren	x			
Jackson, Carolyn		x		
Jacobs, Jonathan L.	x			
Jung, Sook	x			
Kauwe, Angela N.	x			
Kethers, Stefanie	x			
Kumari, Sunita		x		
Lupu, Viorica	x			
Main, Dorrie	x	x		
Marrano, Annarita	x			
Mason, Charles J.	x			
Mayo, Keith				Editorial relationship: Cyberbiosecurity Quarterly
McKenna, Duane		x		
McNamara, John		x		
Meyers, Natalie			x	
Molla Morales, Almudena				Personal
Mungall, Christopher J.	x			
Naithani, Sushma		x		
Nayduch, Dana	x			
Nicholas, Guise				Editorial relationship: Cyberbiosecurity Quarterly
Novikova, Olga S.	x			
Olds, Cassandra	x			
Otto, Ryan		x		
Perkins, Alex			x	
Pesce, Valeria	x			
Pfrender, Michael			x	
Pickens, Victoria	x			
Pitzer, Jimmy	x			
Poelchau, Monica		x		
Provin, Ellen	x			
Puzis, Rami				Editorial relationship: IEEE cyberbio26 Program Committee
Raciti, Daniela	x			
Reid, Bill				Editorial relationship: Cyberbiosecurity Quarterly
Reiser, Leonore	x	x		
Richards, Stephen	x			

Roche, Benjamin				Personal
Scheffler, Brian E.	x			
Schrader, Jeremy E.	x			
Scroggs, Stacey L. P.	x			
Selby, Peter		x		
Self, Adrian				Editorial relationship: Cyberbiosecurity Quarterly
Sen, Taner		x		
Sharma, Sharu P.	x			
Shults, Phillip	x			
Sibley, Gwendolyn		x		
Silver, Kristopher	x			
Sim, Sheina B.	x			
Simmonds, Tyler J.	x			
Stahlke, Amanda R.	x			
Stein, Lincoln	x			
Sternberg, Paul W.	x			
Taylor, Caitlin	x			
Tello-Ruiz, Marcela Karey		x		
Vasserman, Eugine				Editorial relationship: Cyberbiosecurity Quarterly
Wilkinson, Mark D.	x			
Wright, Adam J.	x			
Yook, Karen J.	x			
Zhang, Huiting	x			
Zimmer, Andrew				Family
Zimmer, Margaret				Family

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CONFLICT OF INTEREST LIST

Name: Gwendolyn Sibley

Instructions:

- Who completes this template: Each project director/principal investigator (PD/PI) or other person that the Request for Applications (RFA) specifies
- How this template is completed:
  - List alphabetically – with last name first -- the full names of the following individuals:
    - All co-authors on publications within the past three years, including pending publications and submissions
    - All collaborators on projects within the past three years, including current and planned collaborations
    - All thesis or postdoctoral *advisees/advisors*
    - All persons in your field with whom you have had a consulting/financial arrangement/other conflict-of-interest in the past three years
  - Indicate the person's relationship to you (Co-Author, Collaborator, etc) with an "x".

Additional pages may be used as necessary.

Note: Other individuals working in the applicant's specific area are not in conflict of interest with the applicant unless those individuals fall within one of the listed categories.

Name	Co-Author	Collaborator	Advisees/ Advisors	Other – Specify Nature
Anderson, Alice		X		
Andresen, Dan	X			
Berney, Elizabeth		X		
Brooks-Kieffer, Jamene		X		
Coleman, Jason		X		
Crawley, Francis	X			
Davis, Makay	X			Mentor
Deiser, Raymond		X		
Durham, E		X		
Glymour, Bruce		X		
Hale, Raelynne	X	X		
Hall, Danielle	X			
Hitzler, Pascal		X		
Jackson, Carolyn	X	X		
Kickbusch, Scott	X			
McCowen, Robert	X			
Meyers, Natalie	X			
Misilo, Tom	X	X		
Mocnik, Joe	X			
Molik, David	X	X	X	Supervisor
Olds, Cassandra		X		
Olsen, Livia		X		
Otto, Ryan	X	X		
Peck, Alex	X			Mentor

Schultz, Tristan (Nova)	X			Supervised
Sevin, Carol		X		
Shippy, Teresa		X		
Slimp, Mickey		X		
Spahr, Kendra		X		
Tank, William		X		
Turvey-Welch, Michelle	X			
Vasserman, Eugene		X		
Van Ittersum, Christina		X		
Wilkinson, Mark	X			
Wiser, James	X	X		
Wright, Adam	X			

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